Sparse Contamination in Tail-Index Estimation: Detectability, Negligibility, and Risk

Diogo Ribeiro

Faculty of Media Arts and Design, Technical University of Porto, Porto, Portugal

ORCID 0009-0001-2022-7072

Corresponding author: dfr@esmad.ipp.pt

August 29, 2026

Abstract

Robust tail-index estimators are usually judged by whether they identify the contaminated extremes. This paper separates that diagnostic target from two decision targets: mean-squared-error (MSE) worthwhile intervention and first-order risk relevance. In an exact Pareto model, multiplying the largest r_n observations by Δ_n shifts exactly one normalized log-spacing, by $S_n = r_n \log \Delta_n$. Three thresholds in S_n govern three questions. Trimmed-Hill oracle intervention repays its variance cost above $\gamma\{r_n k_n/(k_n - r_n)\}^{1/2}$; the conservative spacing scan detects the contaminated boundary above $\gamma \log(h_n/\alpha_n)$ for scan size h_n and level α_n; and ordinary Hill risk is disturbed at first order only at $\gamma\sqrt{k_n}$. For bounded contamination counts these are ordered

$$\gamma\sqrt{r_n k_n/(k_n - r_n)} \ll \gamma \log(h_n/\alpha_n) \ll \gamma\sqrt{k_n},$$

so the three are distinct. The main first-order result is a detect-or-ignore theorem: strong contamination is detected and removed, while weak contamination can be missed without changing the first-order limit distribution. A pre-specified Monte Carlo experiment of 5000 paired replications reproduces the two exact risk identities and exhibits the predicted detection transition, and bounded-influence estimators outperform adaptive trimming in part of the region where intervention is worthwhile but recovery is unreliable. Post-specified stress tests bound the detection scale's scope: contamination spread across several spacings damages Hill identically while remaining nearly undetectable. For ordered unequal factors, Hill damage is governed by the sum of the local spacing shifts and scan detectability by their largest component, so localization rather than total magnitude is the relevant diagnostic quantity.

Keywords extreme value theory · tail index · Hill estimator · robust estimation · adaptive trimming · sparse contamination

Mathematics Subject Classification 62G32 · 62G35 · 62G20

1 Question

Recovering the contaminated observations and controlling squared-error risk are different objectives, and it is not obvious that the first is needed for the second:

does a robust estimator need to recover every contaminated extreme to have good risk?

The analysis below shows that the answer is no, at least in the structured exact-Pareto contamination model studied here. Exact recovery is a diagnostic target. Squared-error risk is a decision target. Sparse contamination can be worth removing for an oracle, too weak for a conservative detector to recover reliably, and still too weak to matter at the ordinary first-order sampling-error scale.

The model is deliberately structured extremal contamination: the top r_n clean order statistics are modified after their ranks are known. This is a stress-test model for tail robustness, not generic Huber contamination and not randomly placed measurement error.

The key signal in the equal-strength model is

$$S_n = r_n \log \Delta_n,$$

where r_n is the number of top observations perturbed and Δ_n is their multiplicative strength. The same signal appears in the exact MSE gain from oracle trimming, in spacing-test detectability and in Hill bias. The point of the paper is that those appearances use different normalizing scales.

2 Exact Pareto Spacing Model

Let $X_1, \ldots, X_n$ be exact Pareto with survival function

$$\overline{F}(x) = x^{-1/\gamma}, \qquad x \geq 1, \quad \gamma > 0.$$

Write the descending order statistics as

$$X_{1:n} \geq X_{2:n} \geq \cdots \geq X_{n:n}.$$

For $j = 1, \ldots, k_n$, define the normalized upper log-spacings

$$Z_j = j\{\log X_{j:n} - \log X_{j+1:n}\}.$$

By the exponential spacing representation (Renyi, 1953), $Z_1, \ldots, Z_{k_n}$ are independent exponential random variables with mean γ. This exactness is specific to the Pareto model, but the log-spacing representation itself is not: for Pareto-type tails the same spacings admit an approximate exponential representation with a second-order remainder (Beirlant et al., 2002), which is the sense in which the calculations below are a controlled benchmark rather than an isolated special case.

The ordinary Hill estimator at threshold k_n is

$$\widehat{\gamma}_H(k_n) = \frac{1}{k_n} \sum_{j=1}^{k_n} Z_j.$$

The fixed-r trimmed Hill estimator is

$$\widehat{\gamma}_T(r, k_n) = \frac{1}{k_n - r} \sum_{j=r+1}^{k_n} Z_j.$$

3 Sparse Multiplicative Contamination

For integers $0 \leq r_n < k_n$, define the contaminated ordered sample by

$$X^{\star}_{j:n} = \begin{cases} \Delta_n X_{j:n}, & 1 \leq j \leq r_n, \\ X_{j:n}, & j > r_n, \end{cases} \qquad \Delta_n \geq 1.$$

Multiplying the top r_n clean order statistics by the same factor preserves their internal log-gaps. It changes only the boundary gap below the last perturbed observation.

Lemma 1 (Boundary-spacing identity)**.** *Let*

$$Z_j^\star = j\{\log X_{j:n}^\star - \log X_{j+1:n}^\star\}.$$

If $r_n \geq 1$, then

$$Z_j^\star = Z_j \quad (j \neq r_n), \qquad Z_{r_n}^\star = Z_{r_n} + r_n \log \Delta_n.$$

Consequently,

$$\widehat{\gamma}_H^\star(k_n) - \widehat{\gamma}_H(k_n) = \frac{S_n}{k_n}, \qquad S_n = r_n \log \Delta_n.$$

Proof. For $j < r_n$, both $X_{j:n}$ and $X_{j+1:n}$ are multiplied by Δ_n, so the log-ratio is unchanged. For $j > r_n$, neither value is multiplied. At $j = r_n$, only the upper value is multiplied, so the log-ratio increases by

$$\log(\Delta_n X_{r_n:n}) - \log X_{r_n+1:n} - \{\log X_{r_n:n} - \log X_{r_n+1:n}\} = \log \Delta_n.$$

Multiplication by the spacing index r_n gives the displayed identity. The Hill identity follows by averaging the first k_n normalized spacings. □

Remark 1. *If $r_n = 0$, then $S_n = 0$ and the identities are read with no contaminated boundary. The equal-factor model is the cleanest version of the sparse-contamination question because detection and Hill bias collapse to one scalar signal.*

Lemma 2 (Unequal multiplicative factors)**.** *Let the top r clean order statistics be multiplied by positive factors*

$$\Delta_1 \geq \Delta_2 \geq \cdots \geq \Delta_r \geq 1, \qquad \Delta_{r+1} = 1,$$

so that their order is preserved. Then, for $j \leq r$,

$$Z_j^\star = Z_j + j \log \frac{\Delta_j}{\Delta_{j+1}},$$

and $Z_j^\star = Z_j$ for $j > r$. Consequently,

$$\widehat{\gamma}_H^\star(k) - \widehat{\gamma}_H(k) = \frac{1}{k} \sum_{j=1}^{r} \log \Delta_j.$$

Proof. For $j \leq r$, subtracting the contaminated and clean log-gaps gives

$$\{\log(\Delta_j X_{j:n}) - \log(\Delta_{j+1} X_{j+1:n})\} - \{\log X_{j:n} - \log X_{j+1:n}\} = \log(\Delta_j/\Delta_{j+1}).$$

Multiplication by j gives the spacing identity. The Hill shift is

$$\frac{1}{k} \sum_{j=1}^{r} j \log(\Delta_j/\Delta_{j+1}) = \frac{1}{k} \sum_{j=1}^{r} \log \Delta_j,$$

by summation by parts, using $\Delta_{r+1} = 1$. □

Remark 2. *Outside the equal-factor model, detection and Hill risk do not depend on the same scalar. Detection responds to local spacing jumps $j \log(\Delta_j/\Delta_{j+1})$, while Hill bias responds to the cumulative sum $\sum_{j=1}^{r} \log \Delta_j$.*

Remark 3. *Collect those jumps into a vector. Write*

$$\delta_j = Z_j^\star - Z_j = j \log \frac{\Delta_j}{\Delta_{j+1}}, \qquad j = 1, \ldots, r,$$

with $\Delta_{r+1} = 1$, *and* $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_r)$. *The ordering* $\Delta_1 \geq \cdots \geq \Delta_r \geq 1$ *makes* $\delta_j \geq 0$, *and the summation by parts in the proof of Lemma 2 says exactly that*

$$\sum_{j=1}^{r} \delta_j = \sum_{j=1}^{r} \log \Delta_j,$$

so the Hill shift is the ℓ^1 *norm of the shift vector,*

$$\widehat{\gamma}_H^\star(k) - \widehat{\gamma}_H(k) = \frac{\|\boldsymbol{\delta}\|_1}{k}.$$

In the equal-factor case only the boundary spacing moves, $\boldsymbol{\delta} = (0, \ldots, 0, S)$ *with* $S = r \log \Delta$, *so* $\|\boldsymbol{\delta}\|_1 = S$ *and the scalar notation used throughout the rest of the paper is recovered.*

4 Detection Scale

The adaptive trimming rule in the `heavytails` package tests normalized spacings against the mean of deeper spacings. For a candidate boundary j, write

$$R_j = \frac{Z_j}{\overline{Z}_{j+1:k_n}}, \qquad \overline{Z}_{j+1:k_n} = \frac{1}{k_n - j} \sum_{\ell=j+1}^{k_n} Z_\ell.$$

Under the exact Pareto null,

$$\mathbb{P}_0(R_j > t) = \left(\frac{m}{m+t} \right)^m, \qquad m = k_n - j.$$

Testing h_n possible trim boundaries at family-wise level α_n gives the Bonferroni pointwise level

$$a_n = \alpha_n / h_n.$$

The rejection threshold is therefore

$$t_{m,a_n} = m\{a_n^{-1/m} - 1\}.$$

If $\log(1/a_n) = o(m)$, then

$$t_{m,a_n} = \log(1/a_n)\{1 + o(1)\} = \log(h_n/\alpha_n)\{1 + o(1)\}.$$

Proposition 1 (Boundary detectability)**.** *Assume* $1 \leq r_n \leq h_n < k_n$, $m_n = k_n - r_n \to \infty$, $\alpha_n \to 0$ *and* $\log(h_n/\alpha_n) = o(m_n)$. *Let*

$$L_n = \log(h_n/\alpha_n).$$

Under the sparse contamination model of Lemma 1, the boundary test at $j = r_n$ *satisfies:*

$$\frac{S_n}{\gamma L_n} \geq 1 + \varepsilon \quad \Longrightarrow \quad \mathbb{P}(\textit{boundary detected}) \to 1,$$

and

$$\frac{S_n}{\gamma L_n} \leq 1 - \varepsilon \quad \Longrightarrow \quad \mathbb{P}(\textit{boundary detected}) \to 0,$$

for every fixed $\varepsilon > 0$.

Proof. At the contaminated boundary,

$$R^\star_{r_n} = \frac{Z_{r_n} + S_n}{\overline{Z}_{r_n+1:k_n}}.$$

The denominator converges in probability to γ by the law of large numbers, since $m_n \to \infty$. Also $Z_{r_n} = O_p(1)$, while $L_n \to \infty$. Hence

$$\frac{R^\star_{r_n}}{L_n} = \frac{S_n}{\gamma L_n} + o_p(1).$$

The exact rejection threshold obeys $t_{m_n,a_n}/L_n \to 1$. The two claims follow by comparison with this threshold. □

Theorem 1 (Exact recovery by deepest-first scan)**.** *In the equal-factor contamination model, define the deepest-first scan by*

$$\widehat{r}_n = \max\{j \le h_n : p_j \le \alpha_n/h_n\},$$

with $\widehat{r}_n = 0$ *if the set is empty, where* p_j *is the exact spacing p-value. Assume* $1 \le r_n \le h_n < k_n$*,* $k_n - r_n \to \infty$*,* $\alpha_n \to 0$*,* $\log(h_n/\alpha_n) = o(k_n - r_n)$ *and, for some* $\varepsilon > 0$*,*

$$S_n \ge (1+\varepsilon)\gamma \log(h_n/\alpha_n).$$

Then

$$\mathbb{P}(\widehat{r}_n = r_n) \to 1.$$

Proof. By Proposition 1, the contaminated boundary $j = r_n$ is rejected with probability tending to one. If no deeper clean spacing $j > r_n$ is falsely rejected, the maximum rejected index is therefore exactly r_n. For every $j > r_n$, the spacing and all deeper spacings are unchanged exact-Pareto spacings, so the corresponding null p-value is uniform. A union bound gives

$$\mathbb{P}\{\exists j > r_n : p_j \le \alpha_n/h_n\} \le (h_n - r_n)\frac{\alpha_n}{h_n} \le \alpha_n \to 0.$$

Combining boundary detection with the vanishing probability of a deeper false rejection proves the result. □

Exact recovery has the same leading boundary as detecting the contaminated boundary, provided the boundary is inside the scan and the equal-factor model holds. This is stronger than detecting some anomaly, but it is still a detection statement rather than a risk statement.

5 First-Order Hill Risk Scale

Under the clean exact Pareto model,

$$\sqrt{k_n}\{\widehat{\gamma}_H(k_n) - \gamma\} \Rightarrow N(0, \gamma^2).$$

Lemma 1 gives the contaminated estimator

$$\sqrt{k_n}\{\widehat{\gamma}^\star_H(k_n) - \gamma\} = \sqrt{k_n}\{\widehat{\gamma}_H(k_n) - \gamma\} + \frac{S_n}{\sqrt{k_n}}.$$

Therefore the contamination is negligible on the ordinary sampling-error scale when

$$S_n = o(\gamma\sqrt{k_n}),$$

and first-order relevant when

$$S_n \asymp \gamma\sqrt{k_n}.$$

It dominates ordinary sampling error when

$$S_n/(\gamma\sqrt{k_n}) \to \infty.$$

Proposition 2 (Hill risk shift)**.** *For fixed k_n under exact Pareto sampling and the contamination model above,*

$$\mathrm{E}\{\widehat{\gamma}_H^{\star}(k_n) - \gamma\} = \frac{S_n}{k_n}, \qquad \mathrm{Var}\{\widehat{\gamma}_H^{\star}(k_n)\} = \frac{\gamma^2}{k_n}.$$

Thus

$$\mathrm{MSE}\{\widehat{\gamma}_H^{\star}(k_n)\} = \frac{\gamma^2}{k_n} + \frac{S_n^2}{k_n^2}.$$

Proof. The contaminated estimator is the clean Hill estimator plus the deterministic shift S_n/k_n. The clean Hill estimator is unbiased with variance γ^2/k_n because it averages k_n independent exponential spacings with mean γ and variance γ^2. □

6 Trimmed-Hill Intervention Scale

The $\sqrt{k_n}$ scale is a first-order scale. It is not the exact threshold at which robustification begins to improve MSE. If an oracle knows the true contamination count r_n, it can remove the shifted boundary spacing entirely by using $\widehat{\gamma}_T(r_n, k_n)$.

Proposition 3 (Oracle trimming intervention threshold)**.** *Under the equal-factor contamination model with $0 < r < k$, the oracle trimmed Hill estimator satisfies*

$$\mathrm{MSE}\{\widehat{\gamma}_T(r,k)\} = \frac{\gamma^2}{k-r}.$$

Compared with the contaminated Hill estimator,

$$\mathrm{MSE}\{\widehat{\gamma}_T(r,k)\} - \mathrm{MSE}\{\widehat{\gamma}_H^{\star}(k)\} = \frac{\gamma^2 r}{k(k-r)} - \frac{S^2}{k^2},$$

where $S = r\log\Delta$. Therefore oracle trimming improves exact MSE if and only if

$$S^2 > \gamma^2\frac{rk}{k-r}.$$

The exact-MSE intervention threshold is

$$I_{r,k}^T = \gamma\sqrt{\frac{rk}{k-r}},$$

and, if $r = o(k)$,

$$I_{r,k}^T \sim \gamma\sqrt{r}.$$

Proof. After trimming the first r spacings, the shifted boundary spacing $Z_r^{\star}$ is absent. The remaining $k-r$ spacings are unchanged independent exponentials with mean γ and variance γ^2. Hence the oracle trimmed estimator is unbiased and has variance $\gamma^2/(k-r)$. Proposition 2 gives the contaminated Hill MSE. Subtracting the two MSEs gives

$$\frac{\gamma^2}{k-r} - \left(\frac{\gamma^2}{k} + \frac{S^2}{k^2}\right) = \frac{\gamma^2 r}{k(k-r)} - \frac{S^2}{k^2}.$$

The displayed intervention condition and threshold follow by rearrangement. □

Remark 4. *Any nonzero S changes the exact contaminated-Hill MSE. Proposition 3 asks a sharper decision question: when does oracle robustification compensate for the variance cost of discarding r spacings?*

Remark 5. *The oracle here knows the true trim count within the standard trimmed-Hill family. It is not an oracle over arbitrary estimators, and $I^T_{r,k}$ is correspondingly the intervention scale of that family rather than a universal one. Under the equal-factor model only the boundary spacing Z_r is shifted, by Lemma 1, so a procedure that knew which spacing had moved could discard Z_r alone and keep $Z_1, \dots, Z_{r-1}$. That estimator has MSE $\gamma^2/(k-1)$, so it improves on contaminated Hill as soon as*

$$S^2 > \gamma^2 \frac{k}{k-1}, \qquad \text{that is} \qquad S \gtrsim \gamma,$$

with no dependence on r at all. The growth $I^T_{r,k} \sim \gamma\sqrt{r}$ is therefore entirely the price of discarding $r-1$ uncontaminated spacings, not a property of the contamination: $I^T_{r,k}$ measures when a particular robust action becomes worthwhile, not when intervention as such does. The three-scale ordering is unaffected, because both thresholds are $O(1)$ for fixed r.

7 Detect or Safely Ignore

The three-scale picture has a first-order consequence. An adaptive trimming rule need not recover every weak contaminated boundary to have the same first-order behavior as clean Hill. It is enough that strong contamination is detected and weak contamination is asymptotically invisible at the $k_n^{-1/2}$ scale.

Theorem 2 (Detect-or-ignore first-order robustness)**.** *Consider the equal-factor structured contamination model. Assume*

$$r_n = O(1), \qquad r_n \le h_n < k_n, \qquad k_n \to \infty,$$

$$\alpha_n \to 0, \qquad \log(h_n/\alpha_n) = o(k_n - r_n),$$

and define

$$D_n = \gamma \log(h_n/\alpha_n), \qquad D_n = o(\gamma\sqrt{k_n}).$$

Let $\widehat{r}_n$ be the deepest-first scan from Theorem 1, and define

$$\widehat{\gamma}_A = \widehat{\gamma}_T(\widehat{r}_n, k_n).$$

Then, for every deterministic sequence $\Delta_n \ge 1$, equivalently $S_n = r_n \log \Delta_n \ge 0$,

$$\sqrt{k_n}(\widehat{\gamma}_A - \gamma) \Rightarrow N(0, \gamma^2).$$

Proof. Let $L_n = \log(h_n/\alpha_n)$, so $D_n = \gamma L_n$. The proof uses the subsequence principle.

First take a subsequence along which $S_n > 2D_n$ eventually. Then $S_n \ge (1+\varepsilon)\gamma L_n$ with $\varepsilon = 1$, and Theorem 1 gives $\mathbb{P}(\widehat{r}_n = r_n) \to 1$. On that event, $\widehat{\gamma}_A$ is the oracle trimmed estimator. Since $r_n = O(1)$,

$$\mathrm{Var}\{\sqrt{k_n}(\widehat{\gamma}_T(r_n, k_n) - \gamma)\} = \frac{k_n\gamma^2}{k_n - r_n} \to \gamma^2,$$

and the central limit theorem for the remaining independent exponential spacings gives the displayed limit.

Now take a subsequence along which $S_n \le 2D_n$ eventually. Because $D_n = o(\gamma\sqrt{k_n})$,

$$\frac{S_n}{\sqrt{k_n}} \to 0.$$

With probability at least $1-\alpha_n+o(1)$, the deepest-first scan does not trim past the true boundary: the only way to have $\widehat{r}_n > r_n$ is a false rejection among clean deeper spacings, whose probability is bounded by α_n as in Theorem 1. On the event $\widehat{r}_n \le r_n$, the adaptive estimator differs from the clean Hill estimator by

$$\widehat{\gamma}_A - \widehat{\gamma}_H = O_p(k_n^{-1}) + \mathbf{1}\{\widehat{r}_n < r_n\}\frac{S_n}{k_n - \widehat{r}_n}.$$

The first term is the effect of removing at most $r_n = O(1)$ ordinary exponential spacings and renormalizing by $k_n - \widehat{r}_n$ rather than k_n. The second term is the retained boundary shift when the scan trims too little. Since $S_n \le 2D_n = o(\gamma\sqrt{k_n})$,

$$\sqrt{k_n}(\widehat{\gamma}_A - \widehat{\gamma}_H) = o_p(1),$$

where $\widehat{\gamma}_H$ is the clean Hill estimator built from the unperturbed spacings. The clean Hill central limit theorem gives the same limit.

Every subsequence contains a further subsequence on which either $S_n > 2D_n$ eventually or $S_n \le 2D_n$ eventually. Both further subsequences converge to the same $N(0,\gamma^2)$ law. Therefore the full sequence converges to that law. $\square$

Corollary 1 (Interlock false refusals vanish inside the scan)**.** *Suppose the production deep-contamination interlock scans clean spacings below the supported trimming envelope at family-wise level*

$$\delta_n = \min\{10^{-4}, n^{-2}\}.$$

If the true boundary satisfies $r_n \le h_n$, then

$$\mathbb{P}(\textit{false interlock refusal}) \le \delta_n \to 0.$$

Thus the interlock does not change Theorem 2 when the true contamination boundary lies inside the declared trimming envelope.

Proof. Below the supported trimming envelope the spacings are clean exact-Pareto spacings under the theorem's model. Each interlock p-value is uniform under the null, and the implementation divides the family-wise level across the deep tests. A union bound gives the result. $\square$

Corollary 2 (Missed oracle gains below detection are second-order)**.** *Let*

$$G_n = \mathrm{MSE}\{\widehat{\gamma}_H^{\star}(k_n)\} - \mathrm{MSE}\{\widehat{\gamma}_T(r_n,k_n)\}.$$

If $r_n = O(1)$ and

$$I^T_{r_n,k_n} \ll S_n \ll D_n \ll \gamma\sqrt{k_n},$$

then oracle trimming improves exact MSE, but

$$\frac{G_n}{\gamma^2/k_n} \to 0.$$

Proof. By Proposition 3,

$$G_n = \frac{S_n^2}{k_n^2} - \frac{\gamma^2 r_n}{k_n(k_n - r_n)}.$$

The assumption $S_n \gg I^T_{r_n,k_n}$ makes $G_n > 0$ eventually. Dividing by γ^2/k_n gives

$$\frac{G_n}{\gamma^2/k_n} = \frac{S_n^2}{\gamma^2 k_n} - \frac{r_n}{k_n - r_n}.$$

The first term tends to zero because $S_n \ll \gamma\sqrt{k_n}$, and the second term tends to zero because $r_n = O(1)$ and $k_n \to \infty$. □

8 Separation

The preceding propositions give three different scales:

$$\text{oracle exact-MSE intervention:} \qquad I^T_{r_n,k_n} = \gamma\sqrt{\frac{r_n k_n}{k_n - r_n}},$$

$$\text{detectability:} \qquad D_n = \gamma \log(h_n/\alpha_n),$$

and

$$\text{first-order Hill-risk relevance:} \qquad R_n = \gamma\sqrt{k_n}.$$

Corollary 3 (Three-scale separation)**.** *Suppose $k_n = n^a$ for some $0 < a < 1$, $r_n = O(1)$, $h_n \le C k_n^b$ for fixed $C < \infty$ and $b \ge 0$, and*

$$\alpha_n \asymp (\log n)^{-2}.$$

Then

$$I^T_{r_n,k_n} = O(1), \qquad \log(h_n/\alpha_n) = O(\log n), \qquad \sqrt{k_n} = n^{a/2},$$

so

$$I^T_{r_n,k_n} \ll D_n \ll R_n.$$

There are asymptotic regions in which

$$I^T_{r_n,k_n} \ll S_n \ll D_n \qquad \text{and} \qquad D_n \ll S_n \ll R_n.$$

In the first region oracle trimming improves exact MSE but the conservative adaptive scan does not reliably recover the boundary. In the second region the boundary is detectable but the contamination is negligible at the ordinary first-order Hill sampling-error scale.

The resulting phase diagram is:

signal size	oracle trimming improves MSE?	detectable?	first-order risk relevant?
$S_n \ll I^T_{r_n,k_n}$	no	no	no
$I^T_{r_n,k_n} \ll S_n \ll D_n$	yes	no	no
$D_n \ll S_n \ll R_n$	yes	yes	no
$S_n \gtrsim R_n$	yes	yes	yes

This is the mathematical backbone of this paper. Exact contamination recovery is too demanding as a primary objective, and first-order negligibility is not the same as exact MSE irrelevance. The decision question is when contamination is detectable, when it is worth acting on, and when it alters first-order risk.

9 Estimator Comparison Implied by the Theory

The theory calls for comparing two robustness philosophies rather than ranking estimators.

Detection and removal. Adaptive trimmed Hill estimates r_n and removes the first $\widehat{r}$ normalized spacings. It is naturally targeted at contamination that is detectable by the spacing scan; when such contamination also exceeds the trimmed-Hill intervention scale $I^T_{r,k}$, successful recovery can translate into an MSE benefit. Its vulnerability is the intermediate region $I^T_{r,k} < S < D_n$: oracle trimming would improve exact MSE, but the conservative scan may not recover the boundary. This belongs to the adaptive trimming and extreme-outlier detection line of Bhattacharya, Kallitsis and Stoev (2019); the point here is not to replace that estimator but to separate its recovery target from MSE-worthwhile intervention.

Bounded influence. The t-Hill and harmonic-moment estimators do not try to recover r_n. With threshold $u = X_{k_n+1:n}$, they average powers of reciprocal ratios

$$R_i = u/X_{i:n} \in (0, 1].$$

Sending an observation to infinity drives R_i to zero, so the influence of an individual extreme saturates. These estimators address a different robustness problem: they bound worst-case damage without identifying contaminated observations. This side of the comparison is connected to huberized and bounded-influence robust tail-index estimation (Beran and Schell, 2012; Vandewalle et al., 2007) and to the harmonic-moment and t-Hill literature (Beran et al., 2014; Jordanova et al., 2016).

For harmonic moment parameter $\beta > 0$, define

$$H_\beta = \frac{1}{k}\sum_{i=1}^{k}\left(\frac{u}{X_{i:n}}\right)^\beta, \qquad \widehat{\gamma}_\beta = \frac{1-H_\beta}{\beta H_\beta}.$$

Under equal multiplicative contamination of the top r observations,

$$H^\star_\beta = H_\beta - \frac{1-\Delta^{-\beta}}{k}\sum_{i=1}^{r}\left(\frac{u}{X_{i:n}}\right)^\beta.$$

Thus the contamination effect saturates as $\Delta \to \infty$, unlike Hill's $r \log \Delta/k$ shift.

Proposition 4 (Fixed-sparse harmonic-moment mean contamination shift)**.** *Let $p = \beta\gamma$ and keep r fixed while $k \to \infty$. Under exact Pareto sampling,*

$$\mathrm{E}(H_\beta - H^\star_\beta) = \frac{1-\Delta^{-\beta}}{k}\frac{\Gamma(k+1)}{\Gamma(k+1+p)}\sum_{i=1}^{r}\frac{\Gamma(i+p)}{\Gamma(i)}.$$

Consequently,

$$\mathrm{E}(H_\beta - H^\star_\beta) \sim (1-\Delta^{-\beta})C_{r,p}k^{-(1+p)}, \qquad C_{r,p} = \sum_{i=1}^{r}\frac{\Gamma(i+p)}{\Gamma(i)}.$$

Since

$$g(H) = \frac{1-H}{\beta H}, \qquad H_0 = \frac{1}{1+\beta\gamma},$$

the first-order linearized mean shift is

$$\mathrm{E}\{g'(H_0)(H_\beta^\star - H_\beta)\} \sim \frac{(1+\beta\gamma)^2}{\beta}(1-\Delta^{-\beta})C_{r,\beta\gamma}k^{-(1+\beta\gamma)}.$$

Proof. Conditional on the threshold $u = X_{k+1:n}$ under the exact Pareto model, the exceedance ratios $R_i = u/X_{i:n}$ have the same order-statistic law as k independent variables with distribution $R^\alpha \sim U(0,1)$, where $\alpha = 1/\gamma$. For the ith most extreme exceedance, equivalently the ith smallest reciprocal ratio,

$$\mathrm{E}(R_{(i)}^\beta) = \frac{\Gamma(i+p)\Gamma(k+1)}{\Gamma(i)\Gamma(k+1+p)}.$$

The displayed expectation for $H_\beta - H_\beta^\star$ follows by summing the changed top-r terms. The asymptotic equivalent uses

$$\frac{\Gamma(k+1)}{\Gamma(k+1+p)} \sim k^{-p}.$$

Finally,

$$g'(H) = -\frac{1}{\beta H^2}, \qquad |g'(H_0)| = \frac{(1+\beta\gamma)^2}{\beta},$$

which gives the first-order linearized mean shift after Taylor expansion around H_0. This proposition deliberately records the deterministic mean component of the bounded-influence response; the fixed-r random fluctuation at the same scale is not collapsed to the constant $C_{r,p}$. □

Remark 6. *If $r \to \infty$ while $r/k \to 0$, then*

$$C_{r,p} \sim \frac{r^{1+p}}{1+p}.$$

In that large-r regime Proposition 4 reduces to the simpler scale

$$\widehat{\gamma}_\beta^\star - \widehat{\gamma}_\beta \asymp \frac{1+\beta\gamma}{\beta}(1-\Delta^{-\beta})\left(\frac{r}{k}\right)^{1+\beta\gamma}.$$

This is the natural theoretical comparison with detection-and-removal.

The signal $S = r\log\Delta$ is exact for Hill contamination cost, oracle trimming and the spacing detector. It is not a universal robustness signal: bounded-influence behavior depends separately on r and $1-\Delta^{-\beta}$. Simulations involving t-Hill or harmonic-moment estimators should therefore use r-specific curves or facets rather than pooling cells only because they share the same S.

The empirical comparison should therefore report recovery, detection, and two risk contrasts separately:

$$\mathbb{P}(\widehat{r} = r_n), \qquad \mathbb{P}(\widehat{r} > 0),$$
$$C_n = \mathrm{MSE}(\widehat{\gamma}_H^\star) - \mathrm{MSE}(\widehat{\gamma}_H), \qquad B_n^{(R)} = \mathrm{MSE}(\widehat{\gamma}_H^\star) - \mathrm{MSE}(\widehat{\gamma}_R^\star).$$

The first risk contrast measures contamination cost. The second is a positive-is-good intervention benefit for robust estimator R: $B_n^{(R)} > 0$ means that R improves on contaminated Hill. Recovery, detection, contamination cost and intervention benefit are different quantities.

Remark 7. *The last display in Proposition 4 is the expectation of the linearization, $\mathrm{E}\{g'(H_0)(H_\beta^\star - H_\beta)\}$, and follows from the exact identity above it by multiplication by the constant $g'(H_0)$. It is not a statement about $\mathrm{E}(\widehat{\gamma}_\beta^\star - \widehat{\gamma}_\beta)$: that would need control of the Taylor remainder and of the interaction between the random $H_\beta - H_0$ and the contamination perturbation, which is not established here. The calculation is used below only to indicate how the contamination enters the harmonic-moment family, and the risk comparisons in Section 12 are measured rather than derived from it.*

10 Literature Positioning

This paper does not propose a new robust estimator. Its contribution is the separation

$$\text{detectability} \neq \text{MSE-worthwhile intervention} \neq \text{first-order risk relevance.}$$

The relevant literature can be organized as follows.

strand	role in this paper
classical tail estimation	Hill estimator and log-spacings (Hill, 1975; Renyi, 1953; Beirlant et al., 2002)
robust Pareto efficiency	the price of protection on clean data (Brazauskas and Serfling, 2000, 2001; Finkelstein et al., 2006; Brzezinski, 2016)
spacing-based robust decisions	squared-error and regression rules (Hsieh, 1999; Minkah et al., 2023)
trim-and-detect robustness	adaptive trimming and recovery (Bhattacharya et al., 2019)
bounded-influence robustness	huberized, t-Hill and harmonic-moment (Beran and Schell, 2012; Beran et al., 2014; Jordanova et al., 2016; Vandewalle et al., 2007; Ruckdeschel and Horbenko, 2013)
regular variation theory	background for second-order extensions (de Haan and Ferreira, 2006; de Haan and Stadtmuller, 1996)

Two of those strands bear directly on what is and is not claimed here.

The robust Pareto literature has established for a long time that protection is not free. Brazauskas and Serfling (2000; 2001) study the trade-off between efficiency under clean Pareto data and robustness to upper outliers, both asymptotically and at small sample sizes, and Finkelstein, Tucker and Veeh (2006) make the trade-off explicit through a tuning parameter. That is the same accounting as the clean control of Table 1, where every robust candidate pays a measurable price on uncontaminated data; the contribution here is not to reduce that price but to say when it buys anything. Brzezinski (2016) provides the broad Monte Carlo comparison of robust Pareto tail-index estimators under several contamination schemes, judged by bias and RMSE. The experiment below is deliberately not another entry in that comparison: the estimator set is small and fixed, and the object of study is the three scales that organize when any of the estimators helps, not which one has the lowest risk.

The closest precedents in vocabulary are spacing-based. Hsieh (1999) combines spacing statistics with a squared-error decision rule, and Minkah, de Wet and Ghosh (2023) robustify tail-index estimation directly through an exponential regression model for log-spacings, with influence-function diagnostics under contamination. Both act on the same statistics used here. The quantity being decided differs: in Hsieh it is how much of the tail sample to use, in Minkah et al. it is how to down-weight spacings, and here it is the number of contaminated top order statistics under a structural contamination model. Dupuis and Victoria-Feser (2006) make the related point that robustness is needed in the selection procedure and not only in the estimator; their decision concerns the tail fraction rather than a contamination count, but the separation of a diagnostic step from the action taken on it is the same distinction the three scales formalize.

Against that background the new content can be stated precisely. The detectability and exact-recovery results for the spacing scan, and the bounded-influence and trimming estimators themselves, are taken from the cited work. What is new here is the comparison of three thresholds for the same contamination signal within one model — the exact MSE threshold $I_{r,k}^{T}$ for the trim-count oracle in Proposition 3, the detectability scale D_n, and the first-order risk scale R_n — their ordering in Corollary 3, and the consequence in Theorem 2 that adaptive trimming attains the clean first-order limit for every contamination-strength sequence, including those it fails to recover. We are not aware of a previous statement that separates these three thresholds. The post-specified unequal-factor analysis of Section 13 adds a fourth element: for this spacing scan under bounded-r exact-Pareto contamination with ordered factors, aggregate Hill damage is governed by $\|\boldsymbol{\delta}\|_1$ whereas local detectability is governed by $\|\boldsymbol{\delta}\|_\infty$.

The exact-Pareto calculations above are therefore a controlled benchmark inside the broader regular-variation framework, not a claim that all sparse contamination problems reduce to exact Pareto sampling.

11 Pre-specified Simulation Design

The exact-Pareto theory above is the design constraint for the primary Monte Carlo experiment. Everything in this section was fixed before any Monte Carlo outcome was inspected; Section 12 then reports that frozen design without modifying it. The primary exact-Pareto experiment fixes

$$\gamma = 0.5.$$

Because all three boundaries scale linearly with γ, one value is enough for the primary scale-separation experiment. The grid was:

$$n \in \{10^3, 10^4, 5 \times 10^4\}, \qquad r \in \{1, 3, 5, 10\},$$

The threshold and scan design were fixed before any risk result was seen:

$$k_n = \lfloor n^{1/2} \rfloor, \qquad h_n = 10, \qquad \alpha_n = \min\{0.05, 1/\log^2 n\}.$$

At $n = 1000$, this design is pre-asymptotic: D_n can exceed R_n. That row is labelled as a stress test, while $n = 10^4$ and 5×10^4 test the intended ordering. Other k_n rates are sensitivity checks, not part of the primary design. For the primary values,

n	k_n	α_n	D_n/γ	R_n/γ
1000	31	0.02096	6.168	5.568
10000	100	0.01179	6.743	10.000
50000	223	0.00854	7.065	14.933

so the ordering $D_n < R_n$ is not visible at the smallest stress-test sample.

The contamination grid was frozen in signal space rather than as a common Δ grid. For each (n, r) we computed

$$I^T_{r,k}, \qquad D_n = \gamma \log(h_n/\alpha_n), \qquad R_n = \gamma\sqrt{k_n},$$

and used the signal points

$$S \in \{0, 0.5I^T_{r,k}, 1.5I^T_{r,k}, 0.5D_n, 1.5D_n, 0.5R_n, 1.5R_n\}.$$

The multiplicative contamination strength is then derived as

$$\Delta = \exp(S/r).$$

This makes every cell target the same theoretical locations relative to the three boundaries. Duplicate and near-duplicate signal points were collapsed before outcomes were seen, by a fixed deterministic tolerance rule. The primary x-axis is $S = r \log \Delta$, with results also shown against

$$\frac{S}{I^T_{r,k}}, \qquad \frac{S}{\gamma \log(h_n/\alpha_n)} \qquad \text{and} \qquad \frac{S}{\gamma\sqrt{k_n}}.$$

The estimator set was:

estimator	role
Hill	unprotected reference
oracle trimmed Hill	known-contamination benchmark
adaptive trimmed Hill	detection-and-removal
harmonic moment	bounded influence with $\beta \in \{0.5, 1, 2\}$

The named t-Hill estimator is the $\beta = 1$ harmonic-moment estimator, so it is displayed as t-Hill but is not counted as a separate computational method. Paired Monte Carlo replications were used across all estimators and signal values within each (n, r) cell, the random seed was recorded, and Monte Carlo standard errors for MSE contrasts are reported from the paired replicate losses. The run used a fixed replication count chosen before outcomes were inspected; any larger run would be a precision refinement rather than a design change. The finding the design was built to expose is not that one estimator dominates, but a separation:

$$\text{contamination recovery} \not\Rightarrow \text{MSE-worthwhile intervention},$$

and

$$\text{failure to recover weak contamination} \not\Rightarrow \text{estimator failure}.$$

12 First Exact-Pareto Simulation Run

The first frozen run uses 5000 paired replications with seed 20260826. It records the summary artifact

`research/sparse_contamination/primary_summary.csv`.

The detailed paired-replicate export is intentionally separate from the summary artifact because it is much larger and is only needed for auditing individual replicate losses. Every table below is rendered from those two artifacts by

`research/sparse_contamination/analyze_frozen_run.py`,

which only reads them. The run is frozen and is not re-executed for reporting.

Two reporting conventions are used throughout. First, the design is paired within each (n, r, S) cell, so every MSE contrast is reported as a paired mean together with the Monte Carlo standard error of the replicate loss differences

$$d_i^{(R)} = (\widehat{\gamma}^{\star}_{H,i} - \gamma)^2 - (\widehat{\gamma}^{\star}_{R,i} - \gamma)^2,$$

where R denotes the competing estimator. Because the data-generating process is known, none of these statements is an inferential claim about a population: a quantity is called clearly positive or clearly negative when it lies more than two Monte Carlo standard errors from zero, and comparisons between estimators are read the same way. Second, recovery and detection probabilities are reported with Wilson 95% intervals. Ranges over r describe the design grid only, and are never used as uncertainty statements.

The tables use four estimators on the contaminated sample, written with a star. Alongside contaminated Hill $\widehat{\gamma}^{\star}_H(k_n)$ and the trim-count oracle $\widehat{\gamma}_T(r, k_n)$ of Section 6, write

$$\widehat{\gamma}^{\star}_A = \widehat{\gamma}_T(\widehat{r}, k_n)$$

for the adaptive trimmed Hill estimator, which substitutes the scan's estimate $\widehat{r}$ for the unknown contamination count, and $\widehat{\gamma}^{\star}_{\beta}$ for the harmonic-moment estimator of Section 9 evaluated on the contaminated sample. The oracle is a benchmark rather than a competitor: it is the only one of the four that uses r.

The run was produced with the `heavytails` package version 0.5.0 (Ribeiro, 2026) under Python 3.13.5 and NumPy 2.3.5, at repository commit `755e6ad`. The adaptive trimming scan and the harmonic-moment estimators are that package's implementations, and the cited archive is the same version used here. The run's report artifact records those versions together with the seed and the full configuration. The simulation driver, the analysis-only scripts, the frozen artifacts and that provenance record are collected in a replication package whose manifest carries a SHA-256 checksum for every file. The archive is deposited at Zenodo under `[REPLICATION DOI]`; a compact reproduction guide is supplied as Online Resource 1. Reproducing the numbers reported below requires only the analysis scripts and the frozen artifacts, not a re-run of the simulation.

12.1 Clean control

At $S = 0$ the contamination factor is $\Delta = \exp(S/r) = 1$, so no observation is modified and the true contamination count is $r = 0$ whatever the nominal r of the cell. The four nominal r values therefore describe a single design point drawn four times independently, and they are pooled into the one canonical clean cell of Table 1. The quantity reported there is the correct no-trim rate $\mathbb{P}(\widehat{r} = 0)$, not a trim recovery rate. It is 0.9892 $[0.9877, 0.9905]$ at $n = 10^4$ and 0.9917 $[0.9903, 0.9928]$ at $n = 5 \times 10^4$ over 20000 replications, as expected of a conservative scan run at a vanishing level. Because the sample is uncontaminated here, each estimator's benefit against Hill in that table is exactly the efficiency cost it pays for its robustness, and the costs order as the family's aggressiveness does: at $n = 10^4$ adaptive trimming gives up 0.07×10^{-4} while the harmonic-moment members give up 1.30, 3.67 and 9.64×10^{-4}.

n	reps	$\mathbb{P}(\widehat{r} = 0)$ [95%]	unit	adaptive	β=0.5	β=1	β=2
10^3	20000	0.9768 $[0.9746, 0.9787]$	10^{-3}	-0.26 ± 0.02	-0.59 ± 0.03	-1.53 ± 0.06	-3.89 ± 0.11
10^4	20000	0.9892 $[0.9877, 0.9905]$	10^{-4}	-0.07 ± 0.03	-1.30 ± 0.08	-3.67 ± 0.14	-9.64 ± 0.26
$5{\times}10^4$	20000	0.9917 $[0.9903, 0.9928]$	10^{-4}	-0.01 ± 0.01	-0.54 ± 0.03	-1.54 ± 0.06	-4.02 ± 0.11

Table 1: Clean control. At $S = 0$ the contamination factor is $\Delta = 1$, so the true contamination count is $r = 0$ and the four nominal r values of the frozen grid describe one design point drawn four times independently. The draws are pooled into a single canonical control. The reported rate is the correct no-trim rate $\mathbb{P}(\widehat{r} = 0)$ with a Wilson 95% interval. The remaining columns give each estimator's paired benefit against Hill on this uncontaminated sample, in the unit shown, which is exactly the efficiency cost it pays for its robustness: every value is negative, and they order as the family's aggressiveness does. Oracle trimming is degenerate here and is omitted, since with nothing to remove it coincides with contaminated Hill in every replicate.

12.2 The detection scale

Table 2 reports adaptive trimmed Hill at the two sample sizes where the intended ordering $D_n < R_n$ holds. The transition at D_n is sharp. Below it, at $0.5D$, detection stays low in absolute terms, under 0.052 in every cell, though at about four times the clean-control rate; the benefit is small and negative, because the scan occasionally false-trims and pays for it. Above it, at $1.5D$, detection is 1.0000 in every cell at both sample sizes and the benefit is strongly positive.

n	S	r	S/D_n	detection [95%]	recovery [95%]	$\widehat{B}_n^{(A)}$
10^4	$0.5D$	1	0.500	0.0410 $[0.0358, 0.0469]$	0.0296 $[0.0253, 0.0347]$	$(-2.56 \pm 1.02)10^{-5}$
10^4	$0.5D$	3	0.500	0.0504 $[0.0447, 0.0568]$	0.0386 $[0.0336, 0.0443]$	$(-4.27 \pm 1.14)10^{-5}$
10^4	$0.5D$	5	0.500	0.0456 $[0.0402, 0.0517]$	0.0370 $[0.0321, 0.0426]$	$(-3.79 \pm 1.30)10^{-5}$
10^4	$0.5D$	10	0.500	0.0462 $[0.0407, 0.0524]$	0.0374 $[0.0325, 0.0430]$	$(-2.85 \pm 1.22)10^{-5}$
10^4	$1.5D$	1	1.500	1.0000 $[0.9992, 1.0000]$	0.9892 $[0.9859, 0.9917]$	$(2.41 \pm 0.07)10^{-3}$
10^4	$1.5D$	3	1.500	1.0000 $[0.9992, 1.0000]$	0.9904 $[0.9873, 0.9928]$	$(2.48 \pm 0.07)10^{-3}$
10^4	$1.5D$	5	1.500	1.0000 $[0.9992, 1.0000]$	0.9942 $[0.9917, 0.9960]$	$(2.40 \pm 0.07)10^{-3}$
10^4	$1.5D$	10	1.500	1.0000 $[0.9992, 1.0000]$	1.0000 $[0.9992, 1.0000]$	$(2.29 \pm 0.08)10^{-3}$
10^4	$1.5I{=}0.5R$	10	0.741	0.1870 $[0.1764, 0.1980]$	0.1820 $[0.1716, 0.1929]$	$(-2.03 \pm 0.23)10^{-4}$
10^4	$0.5R$	1	0.741	0.1788 $[0.1684, 0.1897]$	0.1690 $[0.1589, 0.1796]$	$(-1.24 \pm 0.21)10^{-4}$
10^4	$0.5R$	3	0.741	0.1798 $[0.1694, 0.1907]$	0.1680 $[0.1579, 0.1786]$	$(-1.42 \pm 0.21)10^{-4}$
10^4	$0.5R$	5	0.741	0.1778 $[0.1675, 0.1886]$	0.1692 $[0.1591, 0.1798]$	$(-1.62 \pm 0.21)10^{-4}$
10^4	$1.5R$	1	2.224	1.0000 $[0.9992, 1.0000]$	0.9896 $[0.9864, 0.9921]$	$(5.47 \pm 0.11)10^{-3}$
10^4	$1.5R$	3	2.224	1.0000 $[0.9992, 1.0000]$	0.9922 $[0.9894, 0.9943]$	$(5.62 \pm 0.11)10^{-3}$
10^4	$1.5R$	5	2.224	1.0000 $[0.9992, 1.0000]$	0.9944 $[0.9919, 0.9961]$	$(5.51 \pm 0.11)10^{-3}$
10^4	$1.5R$	10	2.224	1.0000 $[0.9992, 1.0000]$	1.0000 $[0.9992, 1.0000]$	$(5.24 \pm 0.11)10^{-3}$
$5{\times}10^4$	$0.5D$	1	0.500	0.0350 $[0.0303, 0.0405]$	0.0296 $[0.0253, 0.0347]$	$(-6.22 \pm 3.17)10^{-6}$
$5{\times}10^4$	$0.5D$	3	0.500	0.0382 $[0.0332, 0.0439]$	0.0302 $[0.0258, 0.0353]$	$(-8.63 \pm 3.22)10^{-6}$
$5{\times}10^4$	$0.5D$	5	0.500	0.0414 $[0.0362, 0.0473]$	0.0336 $[0.0290, 0.0390]$	$(-5.84 \pm 3.43)10^{-6}$
$5{\times}10^4$	$0.5D$	10	0.500	0.0370 $[0.0321, 0.0426]$	0.0302 $[0.0258, 0.0353]$	$(-5.04 \pm 2.94)10^{-6}$
$5{\times}10^4$	$1.5D$	1	1.500	1.0000 $[0.9992, 1.0000]$	0.9922 $[0.9894, 0.9943]$	$(5.59 \pm 0.23)10^{-4}$
$5{\times}10^4$	$1.5D$	3	1.500	1.0000 $[0.9992, 1.0000]$	0.9938 $[0.9912, 0.9956]$	$(5.45 \pm 0.23)10^{-4}$
$5{\times}10^4$	$1.5D$	5	1.500	1.0000 $[0.9992, 1.0000]$	0.9948 $[0.9924, 0.9964]$	$(5.31 \pm 0.23)10^{-4}$
$5{\times}10^4$	$1.5D$	10	1.500	1.0000 $[0.9992, 1.0000]$	1.0000 $[0.9992, 1.0000]$	$(4.91 \pm 0.23)10^{-4}$
$5{\times}10^4$	$0.5R$	1	1.057	0.9414 $[0.9345, 0.9476]$	0.9330 $[0.9257, 0.9396]$	$(1.73 \pm 0.15)10^{-4}$
$5{\times}10^4$	$0.5R$	3	1.057	0.9326 $[0.9253, 0.9392]$	0.9256 $[0.9180, 0.9326]$	$(1.32 \pm 0.15)10^{-4}$
$5{\times}10^4$	$0.5R$	5	1.057	0.9348 $[0.9276, 0.9413]$	0.9310 $[0.9236, 0.9377]$	$(1.12 \pm 0.15)10^{-4}$
$5{\times}10^4$	$0.5R$	10	1.057	0.9338 $[0.9266, 0.9404]$	0.9336 $[0.9264, 0.9402]$	$(1.20 \pm 0.16)10^{-4}$
$5{\times}10^4$	$1.5R$	1	3.170	1.0000 $[0.9992, 1.0000]$	0.9916 $[0.9887, 0.9938]$	$(2.52 \pm 0.05)10^{-3}$
$5{\times}10^4$	$1.5R$	3	3.170	1.0000 $[0.9992, 1.0000]$	0.9946 $[0.9922, 0.9963]$	$(2.40 \pm 0.05)10^{-3}$
$5{\times}10^4$	$1.5R$	5	3.170	1.0000 $[0.9992, 1.0000]$	0.9938 $[0.9912, 0.9956]$	$(2.51 \pm 0.05)10^{-3}$
$5{\times}10^4$	$1.5R$	10	3.170	1.0000 $[0.9992, 1.0000]$	1.0000 $[0.9992, 1.0000]$	$(2.48 \pm 0.05)10^{-3}$

Table 2: Detection transition for adaptive trimmed Hill at the two sample sizes where the intended ordering $D_n < R_n$ holds. Detection is $\mathbb{P}(\widehat{r} > 0)$ and recovery is $\mathbb{P}(\widehat{r} = r)$, each with a Wilson 95% interval. The benefit $\widehat{B}_n^{(A)} = \mathrm{MSE}\{\widehat{\gamma}_H^\star\} - \mathrm{MSE}\{\widehat{\gamma}_A^\star\}$ is reported with the Monte Carlo standard error of the paired replicate losses $d_i = (\widehat{\gamma}_{H,i}^\star - \gamma)^2 - (\widehat{\gamma}_{A,i}^\star - \gamma)^2$; positive values favour adaptive trimming. The clean control is reported separately in Table 1. At $n = 10^4, r = 10$ the deterministic tolerance rule collapses the duplicate signal $1.5I = 0.5R$ into the single cell shown.

The $0.5R$ row isolates the same boundary while holding its label fixed. It lies below D_n at $n = 10^4$, where $S/D_n = 0.741$, and above D_n at $n = 5 \times 10^4$, where $S/D_n = 1.057$; detection moves accordingly from about 0.18 to about 0.93. At $n = 10^4, r = 10$ the deterministic tolerance rule collapses the duplicate signal $1.5I = 0.5R$, so that cell appears once, under its own label.

12.3 The intervention scale

The detection experiment does not test Proposition 3, which concerns a different boundary: the signal at which discarding r spacings begins to repay their variance cost. That boundary is invisible to the scan. Every I cell of the frozen grid lies below the detection scale, with $S/D_n \le 0.741$ at $n \ge 10^4$ and $S/D_n \le 0.934$ including the stress row of Appendix A, so adaptive trimming is largely inactive throughout the I region — detection never exceeds 0.19 there at $n \ge 10^4$ — and the intervention scale can be probed only with the trim-count oracle.

Table 3 does that. The Monte Carlo reproduces the sign change predicted by Proposition 3 in every cell of the frozen grid: the oracle benefit is negative in all twelve cells at $0.5I$, clearly so in ten of them, and positive in all twelve cells at $1.5I$, clearly so in all twelve. Proposition 3 is an exact algebraic identity in this model, so the simulation is not evidence for it. What the table establishes is that the implementation reproduces the proposition's finite-sample prediction, in a region where the scan remains below its detection boundary and reliable removal is unavailable.

n	S	r	S/D_n	detection	$\widehat{B}_n^{(T)}$	t
10^3	$0.5I$	1	0.082	0.0238	$(-2.43 \pm 0.47)10^{-4}$	-5.2
10^3	$0.5I$	3	0.148	0.0196	$(-6.90 \pm 0.86)10^{-4}$	-8.0
10^3	$0.5I$	5	0.198	0.0226	$(-1.11 \pm 0.11)10^{-3}$	-9.8
10^3	$0.5I$	10	0.311	0.0246	$(-2.88 \pm 0.19)10^{-3}$	-14.8
10^3	$1.5I$	1	0.247	0.0298	$(2.93 \pm 0.73)10^{-4}$	4.0
10^3	$1.5I$	3	0.443	0.0476	$(1.06 \pm 0.13)10^{-3}$	7.9
10^3	$1.5I$	5	0.594	0.0896	$(2.11 \pm 0.18)10^{-3}$	11.9
10^3	$1.5I$	10	0.934	0.4134	$(4.71 \pm 0.28)10^{-3}$	16.7
10^4	$0.5I$	1	0.075	0.0124	$(-9.67 \pm 7.92)10^{-6}$	-1.2
10^4	$0.5I$	3	0.130	0.0094	$(-5.62 \pm 1.40)10^{-5}$	-4.0
10^4	$0.5I$	5	0.170	0.0164	$(-7.87 \pm 1.90)10^{-5}$	-4.1
10^4	$0.5I$	10	0.247	0.0150	$(-2.10 \pm 0.26)10^{-4}$	-8.0
10^4	$1.5I$	1	0.224	0.0158	$(2.64 \pm 1.28)10^{-5}$	2.1
10^4	$1.5I$	3	0.391	0.0238	$(8.41 \pm 2.22)10^{-5}$	3.8
10^4	$1.5I$	5	0.510	0.0384	$(1.49 \pm 0.28)10^{-4}$	5.3
10^4	$1.5I{=}0.5R$	10	0.741	0.1870	$(3.26 \pm 0.42)10^{-4}$	7.7
$5{\times}10^4$	$0.5I$	1	0.071	0.0064	$(-4.02 \pm 2.47)10^{-6}$	-1.6
$5{\times}10^4$	$0.5I$	3	0.123	0.0060	$(-1.29 \pm 0.40)10^{-5}$	-3.2
$5{\times}10^4$	$0.5I$	5	0.160	0.0098	$(-1.88 \pm 0.55)10^{-5}$	-3.4
$5{\times}10^4$	$0.5I$	10	0.229	0.0144	$(-3.81 \pm 0.79)10^{-5}$	-4.8
$5{\times}10^4$	$1.5I$	1	0.213	0.0118	$(1.05 \pm 0.37)10^{-5}$	2.8
$5{\times}10^4$	$1.5I$	3	0.370	0.0220	$(1.44 \pm 0.66)10^{-5}$	2.2
$5{\times}10^4$	$1.5I$	5	0.480	0.0304	$(4.04 \pm 0.85)10^{-5}$	4.8
$5{\times}10^4$	$1.5I$	10	0.687	0.1204	$(7.28 \pm 1.26)10^{-5}$	5.8

Table 3: Finite-sample check of Proposition 3. The oracle benefit $\widehat{B}_n^{(T)} = \mathrm{MSE}\{\widehat{\gamma}_H^\star\} - \mathrm{MSE}\{\widehat{\gamma}_T(r,k)\}$ is reported with the Monte Carlo standard error of the paired replicate losses, and t is the corresponding paired statistic. The proposition predicts $\widehat{B}_n^{(T)} < 0$ below $I_{r,k}^T$ and $\widehat{B}_n^{(T)} > 0$ above it. Every I cell of the frozen grid lies below the detection scale, so adaptive detection is nearly inactive throughout this table and the intervention boundary is visible only to the oracle.

12.4 Detection and removal against bounded influence

The two robustness philosophies of Section 9 can now be separated empirically. Table 4 contrasts adaptive trimmed Hill with the whole pre-specified bounded-influence family, $\beta \in \{0.5, 1, 2\}$, and with the oracle, and Table 5 gives the paired test of each β against adaptive trimming. No single β is best across the grid. The ordering inside the family is itself informative: increasing β bounds influence more aggressively but carries a larger clean-sample efficiency cost. Across the whole frozen grid at $n \geq 10^4$, reported in Table 10, $\beta = 0.5$ has the larger benefit than adaptive trimming in 20 of 47 cells, $\beta = 1$ in 5, and $\beta = 2$ in none; the only cells in which $\beta = 2$ wins anywhere in the experiment are four in the pre-asymptotic $n = 10^3$ row. That ranking is an outcome of the run rather than a design choice, and $\beta = 0.5$ is not treated below as though it had been the pre-specified competitor.

The frozen grid splits into three regions, organized by the three scales. Below the intervention scale, at $0.5I$, no intervention is worthwhile: the adaptive estimator correctly does almost nothing and stays the closest of the robust candidates to contaminated Hill, while every bounded-influence member pays a standing efficiency cost and loses, with t from -6.2 to -19.4.

Between the intervention and detection scales — the cells $1.5I$, $0.5D$ and, at $n = 10^4$, $0.5R$, which satisfy $S > I^T_{r,k}$ and $S < D_n$ — oracle trimming would help but the conservative scan cannot see the contamination, and bounded influence can pay. At $n = 10^4$ the $\beta = 0.5$ benefit is positive while the adaptive benefit is negative, with $t = 6.1$, 6.7 and 19.4. The advantage is not uniform. At $n = 5 \times 10^4$ the same comparison gives only $t = 0.3$ at $1.5I$ and $t = 1.6$ at $0.5D$, neither beyond Monte Carlo error, and $\beta = 1$ is ahead of adaptive trimming in only one cell of the band. The defensible statement is that bounded influence *can* have an advantage where detection is unavailable, not that it dominates there.

Above the detection scale the adaptive estimator turns on: recovery approaches one and the benefit approaches the trim-count oracle, matching it to well within Monte Carlo error at $1.5D$ and $1.5R$ and beating all three β there. The change of regime is not sharp at D_n, however. At $n = 5 \times 10^4$ the $0.5R$ cell already sits above the detection scale, at $S/D_n = 1.057$ with detection probability 0.93, and $\beta = 0.5$ is still ahead of adaptive trimming there by $t = 4.8$. D_n is the scale at which the scan becomes reliable. Where adaptive trimming overtakes a particular bounded-influence competitor is a different question, involving that competitor's efficiency cost as well, and the frozen grid puts that point above D_n and dependent on β, r and n.

n	S	S/D_n	unit	adaptive	β=0.5	β=1	β=2	oracle
10^4	$0.5I$	0.170	10^{-4}	-0.22 ± 0.07	-1.26 ± 0.16	-3.67 ± 0.29	-9.58 ± 0.50	-0.79 ± 0.19
10^4	$1.5I$	0.510	10^{-4}	-0.51 ± 0.11	0.67 ± 0.19	-1.13 ± 0.31	-6.84 ± 0.50	1.49 ± 0.28
10^4	$0.5D$	0.500	10^{-4}	-0.38 ± 0.13	0.93 ± 0.19	-0.73 ± 0.31	-6.20 ± 0.50	1.43 ± 0.29
10^4	$1.5D$	1.500	10^{-3}	2.40 ± 0.07	1.87 ± 0.04	2.08 ± 0.06	1.63 ± 0.08	2.40 ± 0.07
10^4	$0.5R$	0.741	10^{-4}	-1.62 ± 0.21	3.04 ± 0.23	1.87 ± 0.36	-3.48 ± 0.55	4.74 ± 0.38
10^4	$1.5R$	2.224	10^{-3}	5.51 ± 0.11	4.45 ± 0.06	5.06 ± 0.09	4.66 ± 0.10	5.51 ± 0.11
5×10^4	$0.5I$	0.160	10^{-4}	-0.01 ± 0.02	-0.48 ± 0.07	-1.47 ± 0.12	-3.98 ± 0.21	-0.19 ± 0.05
5×10^4	$1.5I$	0.480	10^{-4}	-0.02 ± 0.03	0.00 ± 0.08	-0.86 ± 0.13	-3.21 ± 0.22	0.40 ± 0.08
5×10^4	$0.5D$	0.500	10^{-4}	-0.06 ± 0.03	0.07 ± 0.08	-0.81 ± 0.14	-3.24 ± 0.22	0.45 ± 0.09
5×10^4	$1.5D$	1.500	10^{-4}	5.31 ± 0.23	4.26 ± 0.16	3.90 ± 0.22	1.48 ± 0.29	5.32 ± 0.23
5×10^4	$0.5R$	1.057	10^{-4}	1.12 ± 0.15	1.61 ± 0.11	0.91 ± 0.17	-1.56 ± 0.24	2.22 ± 0.16
5×10^4	$1.5R$	3.170	10^{-3}	2.51 ± 0.05	2.26 ± 0.03	2.36 ± 0.04	2.13 ± 0.05	2.50 ± 0.05

Table 4: Detection-and-removal against the full pre-specified bounded-influence family at $r = 5$. Entries are paired benefits against contaminated Hill with Monte Carlo standard errors, each row written in the unit given in its fourth column; positive values beat contaminated Hill. The harmonic-moment columns are the three frozen values of β, with $\beta = 1$ the named t-Hill estimator. Increasing β bounds influence more aggressively but carries a larger clean-sample efficiency cost: $\beta = 2$ is the weakest column almost everywhere, and no single β is best across the grid. Paired tests of each β against adaptive trimming are in Table 5.

n	S	S/D_n	β=0.5	β=1	β=2
10^4	$0.5I$	0.170	−6.2	−12.1	−18.9
10^4	$1.5I$	0.510	6.1	−2.0	−12.7
10^4	$0.5D$	0.500	6.7	−1.2	−11.9
10^4	$1.5D$	1.500	−13.4	−9.3	−16.2
10^4	$0.5R$	0.741	19.4	10.2	−3.6
10^4	$1.5R$	2.224	−20.3	−11.5	−16.6
$5{\times}10^4$	$0.5I$	0.160	−6.8	−12.4	−19.4
$5{\times}10^4$	$1.5I$	0.480	0.3	−6.4	−14.9
$5{\times}10^4$	$0.5D$	0.500	1.6	−5.7	−14.8
$5{\times}10^4$	$1.5D$	1.500	−8.9	−11.2	−19.0
$5{\times}10^4$	$0.5R$	1.057	4.8	−1.6	−12.8
$5{\times}10^4$	$1.5R$	3.170	−14.5	−10.6	−17.7

Table 5: Paired comparison of each frozen bounded-influence member against adaptive trimming at $r = 5$. Entries are paired t statistics for the replicate loss differences $d_i = (\widehat{\gamma}^\star_{A,i} - \gamma)^2 - (\widehat{\gamma}^\star_{\beta,i} - \gamma)^2$, so $t > 0$ favours bounded influence and $t < 0$ favours adaptive trimming. The advantages claimed for bounded influence in the undetectable region exceed two Monte Carlo standard errors at some cells and not at others, which is why Section 12 claims only that bounded influence *can* pay there rather than that it dominates.

These regions answer two questions that the detection table alone cannot. Detection and removal pays where the contamination is detectable, and well above the detection boundary, once recovery is effectively complete, adaptive trimming approaches the trim-count oracle. Bounded influence can provide protection in the intermediate region where intervention is worthwhile but reliable detection is unavailable, in the band $I^T_{r,k} < S < D_n$ — the vulnerability identified in Section 9. Neither philosophy dominates the grid, which is the separation Section 11 anticipated.

12.5 The three scales in one view

The pre-specified design required the positive-signal results to be shown against each of the normalized scales $S/I^T_{r,k}$, S/D_n and S/R_n. Figure 1 gives one panel to each, with every positive-signal cell of the frozen grid appearing in all three. Two of the three panels carry an exact prediction rather than only a qualitative transition.

For the intervention scale, Proposition 3 rearranges to

$$\frac{k_n(k_n - r)}{\gamma^2 r}\,[\mathrm{MSE}\{\widehat{\gamma}^\star_H\} - \mathrm{MSE}\{\widehat{\gamma}_T(r, k_n)\}] = \left(\frac{S}{I^T_{r,k}}\right)^2 - 1,$$

so the oracle benefit is one curve in the normalized signal for every (n, r), crossing zero exactly at $I^T_{r,k}$. For the risk scale, Proposition 2 gives the contamination cost to ordinary Hill, in units of the clean Hill MSE γ^2/k_n, as

$$\frac{\mathrm{MSE}\{\widehat{\gamma}^\star_H\} - \mathrm{MSE}\{\widehat{\gamma}_H\}}{\gamma^2/k_n} = \left(\frac{S}{R_n}\right)^2.$$

Both identities hold in the frozen run to Monte Carlo precision. The ratio of observed to predicted Hill MSE inflation has median 1.006, 0.978 and 0.999 at $n = 10^3$, 10^4 and 5×10^4, and the residual of the normalized oracle benefit has median 0.017, −0.071 and −0.034. The detection panel has no closed form and is shown as measured; its transition is the sharpest of the three.

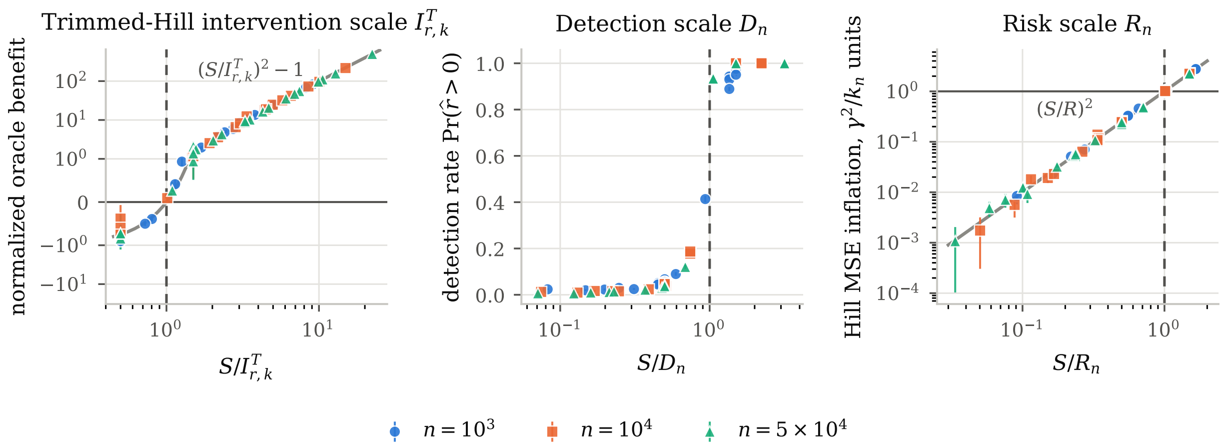

Fig. 1 The frozen run against each of the three normalized scales. Every positive-signal cell of the frozen grid appears in all three panels; the dashed vertical line is that panel's boundary and the grey curve is the exact prediction where one exists. Error bars are Monte Carlo standard errors of the paired replicate losses in the outer panels and Wilson 95% intervals in the detection panel. Each panel's own boundary is the point at which its quantity changes character: the oracle benefit changes sign at $I^T_{r,k}$, the scan transitions around D_n, and the Hill MSE inflation reaches the clean sampling error at R_n. Tables 2–9 give the underlying numbers

The figure also shows the separation itself opening up. The two scales that Corollary 3 orders as $D_n \ll R_n$ have ratio $D_n/R_n = 1.108$ at $n = 10^3$, 0.674 at 10^4 and 0.473 at 5×10^4: the stress-test row is inverted, and the gap widens in the intended direction as n grows. Because the detection panel saturates while the risk panel is still well below one, a contamination that is certainly detected can remain almost irrelevant to first-order Hill risk, which is the content of Theorem 2.

13 Post-Specified Stress Tests

The experiment of Section 12 was pre-specified, and nothing in it is revised here. The three studies below were specified *after* those results were seen, and are reported as post-specified throughout. Each asks how far the three-scale picture depends on a choice that produced it: the vanishing family-wise level, the equal-factor contamination model, and exact Pareto sampling. They use a separate driver, a separate seed and separate artifacts.

13.1 The detection transition and the level regime

The frozen grid places signals only at $0.5D_n$ and $1.5D_n$, which establishes a difference but does not locate a transition. The asymptotic ordering $I^T_{r,k} \ll D_n$ also depends on the vanishing level $\alpha_n = \min\{0.05, 1/\log^2 n\}$: with a fixed α the detection scale $D_n = \gamma \log(h_n/\alpha)$ is constant in n, so it is $O(1)$ like $I^T_{r,k}$ and the two no longer separate asymptotically. That is a real restriction on the theory, and a separate question from the finite-sample behaviour.

Table 6 maps detection on a grid in S/D_n under both regimes. The transition is sharp but centred slightly below one: detection is 0.03–0.08 at $0.4D_n$, has reached 0.73–0.86 at $S = D_n$ itself, and is complete by $S \approx 1.2D_n$. Reporting only $0.5D_n$ and $1.5D_n$ therefore understates how quickly the scan turns on. And the two level regimes give nearly the same curve once each is normalized by its own D_n, the fixed-level detector being uniformly slightly more willing to trim. Over the sample sizes studied the two levels differ little in any case: D_n/γ is 5.298 for fixed $\alpha = 0.05$ against 6.743 and 7.065 for the vanishing level at $n = 10^4$ and 5×10^4, because $\log(1/\alpha_n)$ grows like $2 \log\log n$. The vanishing level is therefore not needed for the finite-sample location of the normalized transition. It is needed for two other things: the asymptotic separation $I^T_{r,k} \ll D_n$ under this design, and the vanishing false-rejection probability that Theorem 1 assumes. With α held fixed the clean false-trim probability does not vanish, so exact recovery is no longer the same statement even where the detection transition sits in the same place.

S/D_n	$n = 10^4$		$n = 5{\times}10^4$	
	vanishing	fixed	vanishing	fixed
0.4	0.027	0.079	0.023	0.084
0.6	0.076	0.152	0.065	0.168
0.7	0.139	0.231	0.129	0.234
0.8	0.269	0.372	0.241	0.378
0.9	0.466	0.574	0.493	0.603
1.0	0.734	0.792	0.810	0.863
1.1	0.909	0.938	0.974	0.986
1.2	0.983	0.990	0.999	1.000
1.3	0.998	0.999	1.000	1.000
1.5	1.000	1.000	1.000	1.000
2.0	1.000	1.000	1.000	1.000

Table 6: Post-specified detection transition at $r = 5$. Entries are $\mathbb{P}(\widehat{r} > 0)$ on a grid in S/D_n, under the vanishing family-wise level $\alpha_n = \min\{0.05, 1/\log^2 n\}$ of the primary design and under a fixed $\alpha = 0.05$. Each column normalizes by its own D_n. The transition is centred slightly below $S/D_n = 1$ and is complete by $S/D_n \approx 1.2$ in both regimes, so the normalized finite-sample transition is similar under fixed and vanishing levels. The vanishing level additionally supplies the asymptotic $I^T_{r,k} \ll D_n$ separation and the vanishing false-rejection probability that Theorem 1 assumes.

13.2 Contamination that is not localized on one spacing

Equal-factor contamination is exceptionally favourable to a spacing scan. By Lemma 1 it moves exactly one normalized spacing, so a problem posed as r contaminated extremes is really a single-spike detection problem, and the scan is built to find exactly that spike. Lemma 2 already says unequal factors behave differently, but the frozen design does not run them.

Table 7 does. Taking $\log \Delta_j = c(r - j + 1)$ puts spacing shifts jc on every $j \leq r$ with the same total, so by Lemma 2 the contaminated-Hill MSE is unchanged: across the cells at $S \geq 1.5D_n$ the two profiles differ in measured contamination cost by at most 5.6%. Detection is not similarly unchanged. At $S = 1.5D_n$ the equal-factor signal is detected in every replicate, while the spread signal is detected in as few as 2.5% of them; at $n = 10^4$ and $r = 10$ the rates are 1.000 against 0.029, and the adaptive benefit falls from 2.39×10^{-3} to 4.55×10^{-5}. The degradation grows with r, which is what dividing a fixed signal among more spacings should do.

The experiment says which contamination the scan misses; the reason is algebraic. Write $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_r)$ for the spacing-shift vector of Remark 3. Contamination enters the scan statistic of Section 4 through both of its arguments:

$$R_j^\star = \frac{Z_j + \delta_j}{\overline{Z}_{j+1:k_n} + \dfrac{1}{k_n - j}\displaystyle\sum_{\ell=j+1}^{r} \delta_\ell},$$

since the deeper spacings that form the denominator are themselves contaminated when $j < r$. The numerator sees one component of $\boldsymbol{\delta}$; the denominator sees the tail of the vector divided by $k_n - j$.

Proposition 5 (Localization-controlled spacing detectability)**.** *Work in the exact-Pareto model with ordered factors $\Delta_1 \geq \cdots \geq \Delta_r \geq 1$, so $\boldsymbol{\delta}_n \geq 0$. Assume $1 \leq r_n \leq h_n < k_n$, $r_n = O(1)$, $\|\boldsymbol{\delta}_n\|_1 = o(k_n)$, $L_n = \log(h_n/\alpha_n) \to \infty$ and $L_n = o(k_n)$, and put*

$$M_n = \|\boldsymbol{\delta}_n\|_\infty = \max_{1 \leq j \leq r_n} \delta_{n,j}.$$

Then, for every fixed $\varepsilon > 0$,

$$\frac{M_n}{\gamma L_n} \geq 1 + \varepsilon \quad \Longrightarrow \quad \mathbb{P}(\textit{some contaminated spacing is rejected}) \to 1,$$

and, for every fixed $\varepsilon \in (0,1)$,

$$\frac{M_n}{\gamma L_n} \leq 1 - \varepsilon \quad \Longrightarrow \quad \mathbb{P}(\textit{any contaminated spacing is rejected}) \to 0.$$

Proof. Write $D_{n,j}$ for the denominator of $R_j^\star$. The contamination it carries is bounded by

$$\frac{1}{k_n - j}\sum_{\ell=j+1}^{r_n} \delta_{n,\ell} \leq \frac{\|\boldsymbol{\delta}_n\|_1}{k_n - r_n} = o(1),$$

because $\|\boldsymbol{\delta}_n\|_1 = o(k_n)$ and $r_n = O(1)$. Since $k_n - j \to \infty$, the law of large numbers gives $\overline{Z}_{j+1:k_n} \to_p \gamma$, so $D_{n,j} \to_p \gamma$; as $r_n = O(1)$ there are boundedly many contaminated indices and the convergence is uniform over $j \leq r_n$. The exact rejection threshold obeys $t_{m_n,a_n}/L_n \to 1$ as in Proposition 1. Because $r_n \leq h_n$, every contaminated spacing is among those the scan tests.

Suppose first $M_n \geq (1+\varepsilon)\gamma L_n$, and let $j^\star \leq r_n$ attain M_n. Under exact Pareto sampling $Z_{j^\star} \geq 0$, so the numerator of $R_{j^\star}^\star$ is at least M_n, and with probability tending to one $D_{n,j^\star} \leq \gamma(1 + \varepsilon/4)$. On that event

$$\frac{R_{j^\star}^\star}{L_n} \geq \frac{M_n}{\gamma(1+\varepsilon/4)L_n} \geq \frac{1+\varepsilon}{1+\varepsilon/4} > 1,$$

a bound that does not require M_n/L_n to stay bounded, and the threshold ratio tends to one, so $j^\star$ is rejected with probability tending to one.

Suppose instead $M_n \leq (1-\varepsilon)\gamma L_n$. Every $\delta_{n,j}/L_n$ is then bounded by $(1-\varepsilon)\gamma$. The contamination in the denominator is nonnegative, so $D_{n,j} \geq \overline{Z}_{j+1:k_n}$, which exceeds $\gamma(1-\varepsilon/2)$ with probability tending to one, and $Z_j/L_n = o_p(1)$ because $Z_j = O_p(1)$ and $L_n \to \infty$. Hence, uniformly over $j \leq r_n$,

$$\frac{R_j^\star}{L_n} \leq \frac{Z_j/L_n + (1-\varepsilon)\gamma}{\gamma(1-\varepsilon/2)} \longrightarrow_p \frac{1-\varepsilon}{1-\varepsilon/2} < 1.$$

A union bound over the boundedly many contaminated indices gives the second claim. $\square$

The proposition concerns the contaminated spacings only. It says nothing about false rejections among the clean spacings, which the level α_n controls separately, and it is therefore not a statement about recovering r_n.

Which quantity matters depends on the question. Hill damage is aggregate: by Remark 3, $\widehat{\gamma}_H^\star - \widehat{\gamma}_H = \|\boldsymbol{\delta}\|_1/k_n$. Detection of *some* contaminated spacing is local, and Proposition 5 puts its boundary at $\|\boldsymbol{\delta}\|_\infty \asymp \gamma L_n = D_n$. Exact recovery of the count is deeper still: the deepest-first scan of Theorem 1 returns $\widehat{r}_n = r_n$ only if the boundary spacing itself is rejected, so the relevant component is δ_r, combined as before with the vanishing false-rejection probability that $\alpha_n \to 0$ supplies. A large $\|\boldsymbol{\delta}\|_\infty$ at some interior $j < r$ does not by itself deliver exact recovery.

For a nonzero, nonnegative $\boldsymbol{\delta} \in \mathbb{R}^r$ define the localization coefficient

$$\kappa(\boldsymbol{\delta}) = \frac{\|\boldsymbol{\delta}\|_\infty}{\|\boldsymbol{\delta}\|_1} \in \left[\frac{1}{r}, 1\right],$$

with $\kappa = 1$ when a single spacing carries the whole shift and $\kappa = 1/r$ when all r carry it equally. The equal-factor model of the primary experiment is the extreme case: there $\boldsymbol{\delta} = (0, \ldots, 0, S)$, so $\kappa = 1$ and $\|\boldsymbol{\delta}\|_1 = \|\boldsymbol{\delta}\|_\infty = S$, which is why one scalar sufficed. In general, writing $S = \|\boldsymbol{\delta}\|_1$ for the aggregate signal, the largest local jump available to the scan is κS, and Proposition 5 places the detection boundary at

$$\kappa S \asymp D_n, \qquad \text{equivalently} \qquad S_{\text{detect}} \asymp \frac{D_n}{\kappa}.$$

This is a localization-adjusted reading of this particular spacing scan under the stated sparse regime, not a detectability boundary for contamination in general or for other detectors.

The spread profile already run is a direct test of that reading, and requires no further simulation. With $\log \Delta_j = c(r - j + 1)$ the shifts are $\delta_j = jc$, so $\|\boldsymbol{\delta}\|_1 = cr(r+1)/2$ and $\|\boldsymbol{\delta}\|_\infty = \delta_r = rc$, giving

$$\kappa = \frac{2}{r+1}.$$

At the $S = 2.5D_n$ cells of Table 7 this predicts local jumps of $\kappa S = 1.25D_n$, $0.833D_n$ and $0.455D_n$ at $r = 3$, 5 and 10, and the measured detection rates fall in that order: 0.994 and 1.000 at $r = 3$, 0.358 and 0.372 at $r = 5$, and 0.060 and 0.055 at $r = 10$, for $n = 10^4$ and 5×10^4. The agreement is quantitative as well as ordinal. Linear interpolation of the equal-factor detection curve of Table 6 in S/D_n, read at $\kappa S/D_n$ instead of S/D_n, predicts 0.991, 0.335 and 0.040 at $n = 10^4$ against the measured 0.994, 0.358 and 0.060; that curve is itself nearly independent of r, agreeing to within 0.01 between $r = 1$ and $r = 5$. The small systematic under-prediction is expected, since the spread profile elevates several spacings at once and the scan tests each of them, while the equal-factor curve describes a single spike of the same height.

This is the sharpest limitation of the paper. Equally damaging contamination is not equally detectable, so D_n is a detection scale for spacing-localized contamination rather than for contamination of a given magnitude.

The consequence for Theorem 2 is stronger than a restriction on D_n: the dichotomy itself does not extend. The theorem holds because contamination is either large enough to be recovered or small enough to be first-order negligible, and under the equal-factor geometry those two cases exhaust the possibilities. Under spread contamination they do not. Since $D_n/R_n = 0.674$ at $n = 10^4$, the cell at $S = 1.5D_n$ already has $S/R_n = 1.011$, so the contamination has reached the first-order Hill-risk scale; with $r = 10$ its detection probability is 0.029 and its recovery probability 0.006. The same happens further out: at $S = 2.5D_n$, where $S/R_n = 1.686$, detection is still only

0.060. Nine cells of the spread design have $S/R_n \geq 1$ and four of them are detected in under a tenth of replications. A contamination can therefore be simultaneously undetected and first-order relevant, which is precisely the case Theorem 2 excludes.

That does not contradict the theorem, which is stated for the equal-factor model and proved there. It does say that the theorem is specific to that geometry rather than to sparse contamination generally, and that a detect-or-ignore statement for unequal factors would need a detection scale built from the local jumps $j\log(\Delta_j/\Delta_{j+1})$ rather than from the total S.

n	r	S/D_n	detection		unit	$\widehat{B}_n^{(A)}$	
			equal	spread		equal	spread
10^4	3	0.5	0.043	0.018	10^{-5}	-5.58 ± 1.03	-2.82 ± 0.95
10^4	3	1.0	0.723	0.052	10^{-5}	9.39 ± 4.02	0.18 ± 1.65
10^4	3	1.5	1.000	0.218	10^{-3}	2.52 ± 0.07	0.08 ± 0.04
10^4	3	2.5	1.000	0.994	10^{-3}	7.00 ± 0.12	6.70 ± 0.12
10^4	5	0.5	0.035	0.020	10^{-5}	-2.48 ± 0.97	-0.81 ± 0.99
10^4	5	1.0	0.733	0.028	10^{-5}	0.52 ± 4.13	-1.20 ± 1.30
10^4	5	1.5	1.000	0.065	10^{-3}	2.37 ± 0.07	0.05 ± 0.02
10^4	5	2.5	1.000	0.358	10^{-3}	6.86 ± 0.12	1.07 ± 0.08
10^4	10	0.5	0.040	0.016	10^{-5}	-2.65 ± 1.19	-0.21 ± 0.80
10^4	10	1.0	0.725	0.021	10^{-5}	-1.31 ± 4.37	1.04 ± 1.02
10^4	10	1.5	1.000	0.029	10^{-3}	2.39 ± 0.08	0.05 ± 0.02
10^4	10	2.5	1.000	0.060	10^{-3}	6.66 ± 0.12	0.30 ± 0.03
5×10^4	3	0.5	0.040	0.014	10^{-5}	-1.33 ± 0.36	0.11 ± 0.25
5×10^4	3	1.0	0.817	0.047	10^{-5}	2.09 ± 1.33	-1.47 ± 0.50
5×10^4	3	1.5	1.000	0.192	10^{-4}	5.55 ± 0.23	0.20 ± 0.12
5×10^4	3	2.5	1.000	1.000	10^{-3}	1.59 ± 0.04	1.59 ± 0.04
5×10^4	5	0.5	0.035	0.014	10^{-6}	-8.63 ± 3.14	-6.05 ± 2.43
5×10^4	5	1.0	0.827	0.025	10^{-5}	2.09 ± 1.39	-0.62 ± 0.34
5×10^4	5	1.5	1.000	0.052	10^{-4}	5.30 ± 0.23	0.16 ± 0.07
5×10^4	5	2.5	1.000	0.372	10^{-3}	1.54 ± 0.04	0.23 ± 0.02
5×10^4	10	0.5	0.040	0.013	10^{-6}	-7.35 ± 3.00	-2.21 ± 1.82
5×10^4	10	1.0	0.808	0.016	10^{-5}	-0.54 ± 1.44	-0.18 ± 0.28
5×10^4	10	1.5	1.000	0.025	10^{-4}	5.05 ± 0.24	0.10 ± 0.05
5×10^4	10	2.5	1.000	0.055	10^{-3}	1.50 ± 0.04	0.07 ± 0.01

Table 7: Post-specified comparison of contamination profiles at matched Hill bias. The equal-factor profile places the whole signal on spacing r; the spread profile uses unequal factors chosen so that, by Lemma 2, the shifts jc fall on all of $j = 1, \ldots, r$ and the total, hence the contaminated-Hill MSE, is unchanged. The two profiles damage Hill identically but are not equally detectable: at $S = 1.5D_n$ the equal-factor signal is detected in every replicate while the spread signal is detected in a few percent of them, and the adaptive benefit falls with it. D_n is therefore a detection scale for spacing-localized contamination, not for contamination of a given magnitude.

13.3 Second-order tails

Under exact Pareto sampling the normalized spacings are exactly iid exponential, which is what makes two of the three scales exact identities. Pareto-type tails admit only an approximate exponential representation (Beirlant et al., 2002), and ordinary Hill then carries threshold bias. Table 8 samples the Burr law

$$\overline{F}(x) = (1 + x^c)^{-\lambda}, \qquad \gamma = \frac{1}{c\lambda}, \qquad \rho = -\frac{1}{\lambda},$$

so that $\gamma = 1/2$ with $\rho \in \{-1, -1/2, -1/4\}$ at $(\lambda, c) = (1, 2), (2, 1), (4, 1/2)$. The top $k_n + 1$ order statistics are generated exactly from uniform order statistics rather than by drawing the full sample; feeding the same generator the exact-Pareto survival quantile returns spacings with mean 0.4992 and variance 0.2496 against the targets $\gamma = 0.5$ and $\gamma^2 = 0.25$, which validates the construction.

The two exact identities of Section 12 cannot survive this change. Lemma 1 is a deterministic identity between spacings, so $\widehat{\gamma}_H^\star = \widehat{\gamma}_H + S/k_n$ whatever law the spacings follow, and therefore

$$\mathrm{MSE}(\widehat{\gamma}_H^\star) - \mathrm{MSE}(\widehat{\gamma}_H) = \frac{S^2}{k_n^2} + \frac{2S}{k_n}\,\mathrm{E}(\widehat{\gamma}_H - \gamma).$$

The second term vanishes only when Hill is unbiased, which is exactly what exact Pareto sampling guarantees. Under a second-order tail it is proportional to the threshold bias, and at $\rho = -1/4$ it is the larger of the two terms. What is tested below is therefore the detection organization, not the risk identities.

Detection stays near its clean level at $0.5D_n$ and is essentially complete at $1.5D_n$ for every ρ, falling only to 0.907 in the most biased cell, and above D_n adaptive trimming approaches the trim-count oracle as recovery becomes nearly complete. What survives is therefore the detection organization; what changes is the risk decomposition, because threshold bias contributes materially to the Hill MSE. At $\rho = -0.25$ and $n = 10^4$ the contaminated-Hill MSE at $1.5D_n$ is 5.4×10^{-2} while the contamination contributes 2.0×10^{-2} of it; the rest is threshold bias. In that regime the contamination question is secondary to the choice of k_n, which the exact-Pareto model removes by construction.

n	ρ	S/D_n	$\mathrm{MSE}(\widehat{\gamma}_H^\star)$	detection	unit	adaptive	oracle
10^4	-1.00	0.0	2.60×10^{-3}	0.013	10^{-4}	-0.11 ± 0.07	-1.30 ± 0.17
10^4	-1.00	0.5	2.92×10^{-3}	0.040	10^{-4}	-0.16 ± 0.11	3.17 ± 0.28
10^4	-1.00	1.5	5.23×10^{-3}	1.000	10^{-3}	2.60 ± 0.07	2.60 ± 0.07
10^4	-0.50	0.0	4.01×10^{-3}	0.008	10^{-4}	0.13 ± 0.06	-2.74 ± 0.23
10^4	-0.50	0.5	5.57×10^{-3}	0.029	10^{-3}	0.03 ± 0.01	1.24 ± 0.03
10^4	-0.50	1.5	1.04×10^{-2}	0.999	10^{-3}	5.95 ± 0.08	5.95 ± 0.08
10^4	-0.25	0.0	3.37×10^{-2}	0.003	10^{-3}	0.03 ± 0.01	-2.13 ± 0.07
10^4	-0.25	0.5	3.98×10^{-2}	0.011	10^{-3}	0.17 ± 0.03	3.95 ± 0.07
10^4	-0.25	1.5	5.44×10^{-2}	0.907	10^{-2}	1.58 ± 0.01	1.79 ± 0.01
5×10^4	-1.00	0.0	1.12×10^{-3}	0.010	10^{-5}	-0.10 ± 0.20	-1.16 ± 0.50
5×10^4	-1.00	0.5	1.21×10^{-3}	0.039	10^{-5}	-0.77 ± 0.33	4.12 ± 0.91
5×10^4	-1.00	1.5	1.69×10^{-3}	1.000	10^{-4}	5.83 ± 0.23	5.83 ± 0.23
5×10^4	-0.50	0.0	1.81×10^{-3}	0.006	10^{-5}	0.25 ± 0.12	-4.15 ± 0.63
5×10^4	-0.50	0.5	2.27×10^{-3}	0.031	10^{-4}	0.22 ± 0.04	3.95 ± 0.10
5×10^4	-0.50	1.5	3.40×10^{-3}	1.000	10^{-3}	1.61 ± 0.02	1.61 ± 0.02
5×10^4	-0.25	0.0	1.91×10^{-2}	0.004	10^{-4}	0.14 ± 0.04	-6.17 ± 0.22
5×10^4	-0.25	0.5	2.15×10^{-2}	0.008	10^{-3}	0.04 ± 0.01	1.54 ± 0.02
5×10^4	-0.25	1.5	2.58×10^{-2}	1.000	10^{-3}	6.24 ± 0.03	6.23 ± 0.03

Table 8: Post-specified second-order stress test at $r = 5$, sampling a Burr law with second-order parameter ρ rather than exact Pareto, so the normalized spacings are only approximately exponential and ordinary Hill carries threshold bias. The detection transition survives: detection stays near its clean level at $0.5D_n$ and is essentially complete at $1.5D_n$ for every ρ, and above D_n adaptive trimming approaches the trim-count oracle as recovery becomes nearly complete. What changes is the relative importance of the contamination: at $\rho = -0.25$ the bias term dominates $\mathrm{MSE}(\widehat{\gamma}_H^\star)$, so the contamination question is secondary to threshold choice.

13.4 What the stress tests change

The stress tests produce two robustness results and one substantive limitation. First, D_n continues to organize the transition under fixed and vanishing family-wise levels. Second, the detection organization persists in the Burr experiments considered here — signals at $0.5D_n$ stay near the clean detection rate, signals at $1.5D_n$ are almost always found, and adaptive trimming approaches the trim-count oracle once recovery is nearly complete — while the exact-Pareto risk identities do not carry over, because threshold bias then contributes materially to the Hill MSE. By contrast, D_n does not characterize detectability for contamination distributed across several spacings. It governs contamination that concentrates on one spacing, and equally damaging contamination spread over several spacings can be nearly invisible to the scan even after it reaches the first-order risk scale. That bounds the reach of Theorem 2 as well as of D_n, and Section 14 treats it as the main open direction rather than as a defect in the separation the paper is about.

14 Discussion

Three scales, not one. A single contamination signal $S_n = r_n \log \Delta_n$ enters three different decisions through three different normalizations, and the frozen experiment separates them. Trim-count-oracle removal changes the sign of its MSE benefit at $I^T_{r,k}$, the conservative spacing scan transitions around D_n, and ordinary Hill risk is disturbed at first order at R_n. Two of the three are exact identities in the exact-Pareto model and the run reproduces both: the Hill MSE inflation matches $(S/R_n)^2$ with median observed-to-predicted ratios 1.006, 0.978 and 0.999 across the three sample sizes, and the normalized oracle benefit matches $(S/I^T_{r,k})^2 - 1$ with median residuals 0.017, -0.071 and -0.034. The detection transition has no closed form and is reported as measured; it is the sharpest of the three.

Within the equal-factor regime, exact recovery is not the decision objective. The detect-or-ignore result of Theorem 2 says that the two ways of failing to recover the contamination are not equally costly, and the primary experiment shows both halves. Within that geometry, above D_n the scan recovers the boundary and adaptive trimming inherits the oracle's benefit. Below D_n it usually does not, but the contamination it misses is then small on the first-order risk scale: among the frozen grid points at $n \geq 10^4$, the largest signal left undetected in more than half of replications has $S/R_n = 0.5$. An estimator can therefore miss contamination without that miss mattering at the scale on which ordinary Hill error is measured. Recovery is a diagnostic target; risk is the decision target, and the two come apart in the band the three scales delimit. Outside the equal-factor geometry the second half of that statement fails, as Section 13 shows: spread contamination can stay undetected past $S/R_n = 1$.

What bounded influence adds. The region where bounded influence can matter is $I^T_{r,k} < S < D_n$, where removal is already worthwhile but the conservative scan cannot reliably find what to remove. There the harmonic-moment estimators limit the contamination's effect without identifying the contaminated boundary, and Table 4 shows that at $n = 10^4$ the $\beta = 0.5$ member has a positive benefit in cells where adaptive trimming has a negative one. The effect is real but not uniform: at $n = 5 \times 10^4$ the same comparison is within Monte Carlo error, and Tables 10 and 11 show that the advantage belongs to particular members rather than to bounded influence as a class. Averaged over the frozen grid at $n \geq 10^4$, $\beta = 0.5$ beats adaptive trimming in 20 of 47 cells, $\beta = 1$ in 5, and $\beta = 2$ in none.

Why no robust method dominates. Each philosophy pays for its protection in a different currency. Trimming pays variance for the spacings it discards, which is why the oracle itself is harmful below $I^T_{r,k}$ and why adaptive trimming is slightly harmful below D_n, where it trims only by mistake. Bounded influence pays a standing efficiency cost that it incurs whether or not contamination is present, which is why every harmonic-moment member is behind adaptive trimming both in the clean control of Table 1, where its cost is an order of magnitude larger than the adaptive estimator's, and in the $0.5I$ cells. Increasing β bounds influence more aggressively but carries a larger clean-sample efficiency cost, so the family is itself a trade-off rather than a ranking. No member is best everywhere, and the location at which adaptive trimming overtakes a given member depends on β, r and n rather than coinciding with D_n.

Two quantities, not one. Proposition 5 replaces the single signal S by two coordinates of the shift vector, and the four regions they define explain why Theorem 2 does not extend beyond the equal-factor model. Writing $x = \|\boldsymbol{\delta}\|_\infty / D_n$ for local detectability and $y = \|\boldsymbol{\delta}\|_1 / R_n$ for first-order Hill relevance,

	$y \lesssim 1$	$y \gtrsim 1$
$x \lesssim 1$	undetected, negligible	**undetected, first-order relevant**
$x \gtrsim 1$	detected, negligible	detected, first-order relevant

Under equal factors $\kappa = 1$, so x and y are the same signal measured against D_n and R_n; with $D_n \ll R_n$ the top-right cell is then empty, which is what makes the detect-or-ignore dichotomy available. Once $\kappa < 1$ the two coordinates separate by the factor κ, and the spread-contamination cells of Section 13 sit in that cell: at $n = 10^4$, $r = 10$ and $S = 1.5D_n$ the aggregate signal has $y = 1.011$ while the largest local jump has $x = 2/11 \times 1.5 = 0.27$. In words, ℓ^1 controls Hill damage and ℓ^∞ controls local detectability; exact count recovery additionally needs the deepest jump δ_r to clear D_n. This describes the spacing scan studied here under the stated sparse regime, not every possible detector.

Limitations. The model is deliberately narrow. Sampling is exact Pareto, so there is no second-order bias term and no bias-variance trade-off in the choice of k_n; the three scales are therefore clean, but real threshold selection is not. The contamination is applied to the top r_n order statistics after their ranks are known and with a common factor, which is a structural stress test rather than Huber contamination. Lemma 2 shows that unequal factors decouple the detection signal from the Hill bias, and Section 13 measures the size of that decoupling: contamination spread over several spacings can damage Hill identically and still be nearly invisible to the scan, and can stay so after reaching the first-order risk scale. That limits Theorem 2 as well as D_n: outside the equal-factor geometry, detection and negligibility no longer exhaust the possibilities, so extending the detect-or-ignore statement to unequal factors is the paper's main open problem rather than a routine generalization. The experiment fixes $\gamma = 0.5$ because all three scales are proportional to γ, uses the single threshold rate $k_n = \lfloor n^{1/2} \rfloor$, and holds r in $\{1, 3, 5, 10\}$, so regimes with $r_n \to \infty$ are untested. It is one frozen run of 5000 replications from one seed; the Monte Carlo standard errors reported throughout quantify that, and the $n = 10^3$ row shows how far the asymptotic ordering can be from a finite sample.

Relation to the robust tail-estimation literature. The contribution is a separation of decision questions rather than a new estimator, so it sits alongside rather than against the trim-and-detect and bounded-influence lines. Robust Pareto estimation has been compared on bias and

RMSE across many estimators and contamination schemes (Brzezinski, 2016), and the efficiency price of protection is long established (Brazauskas and Serfling, 2000; Finkelstein et al., 2006); the clean control here reproduces that price rather than avoiding it. What the three scales add is an answer to a prior question: when the price is worth paying, when the contamination can be found at all, and when it matters to first-order risk. Among spacing-based work, Hsieh (1999) and Minkah, de Wet and Ghosh (2023) act on the same statistics used here, deciding how much tail to use and how to downweight spacings respectively, rather than how many contaminated order statistics to remove.

Open problems. The following extend the results without being prerequisites for them.

1. Extend Proposition 5 into a full unequal-factor detect-or-ignore result, characterizing when local detectability $\|\boldsymbol{\delta}_n\|_\infty$ and aggregate relevance $\|\boldsymbol{\delta}_n\|_1$ together permit or preclude first-order robustness.
2. Extend Theorem 2 to contamination counts with $r_n \to \infty$ but $r_n = o(k_n)$, identifying the growth rates for which finite trimming remains first-order negligible.
3. Convert the harmonic-moment mean-shift calculation into a full MSE comparison with Hill, oracle trimming and adaptive trimming. Section 12 gives that comparison empirically; what is still missing is the analytic counterpart, and in particular a characterization of the risk crossover that Tables 4 and 5 place somewhat above the detection boundary.
4. Compare fixed trimming, adaptive trimming and bounded-influence estimators under the same sparse contamination sequence, extending the finite grid of Section 12 to a sequence in n.
5. Characterize the intervention scale for oracles outside the trimmed-Hill family, following Remark 5: under the equal-factor model a procedure that knew which single spacing was shifted could act on less information loss than r-trimming requires.

15 Conclusion

Robust tail-index estimation is often judged by whether a method finds the contaminated extremes. In the exact-Pareto model studied here, that criterion answers a different question from the one a user of the estimator faces. A sparse multiplicative contamination of the top r_n order statistics enters through one scalar $S_n = r_n \log \Delta_n$, and that scalar has to be compared with three separate scales before anything follows. None of the three is a scale of the contamination alone: each belongs to a procedure. For standard r-trimmed Hill, $I^T_{r,k}$ decides whether removal repays its variance cost; for the conservative spacing scan at level α_n with h_n candidate boundaries, D_n decides whether the contamination can be found at all; and for ordinary Hill under squared error, R_n decides whether the contamination matters at the scale on which its error is measured. For bounded contamination counts r_n these scales are ordered $I^T_{r,k} \ll D_n \ll R_n$, so the intermediate regions are not empty. A pre-specified Monte Carlo experiment reproduces the two exact risk identities at finite n and exhibits the predicted detection transition around D_n.

The practical consequence, within the equal-factor regime of Theorem 2, is that adaptive trimming may safely fail to recover weak contamination, and that below $I^T_{r,k}$ intervention within the standard trimmed-Hill family is not MSE-beneficial, because its variance cost then exceeds the contamination bias it removes. Outside that geometry the two come apart: for ordered unequal factors,

$\|\boldsymbol{\delta}\|_1$ controls Hill's contamination shift while $\|\boldsymbol{\delta}\|_\infty$ controls this spacing scan's local detectability, so contamination can be damaging and undetectable at once. The detection scale determines when reliable removal becomes feasible. The risk comparison between removal and bounding influence depends additionally on the efficiency cost of the bounded-influence procedure, and therefore on β, r and n, rather than on which estimator is more robust in the abstract. Detection, worthwhile intervention, and first-order risk relevance are different statistical questions.

Statements and Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Data and Code Availability

The study uses no empirical data. All results come from the simulations described in Sections 11, 12 and 13. The complete replication package — the simulation drivers, the analysis-only scripts, the frozen summary and per-replicate losses, the configuration and provenance records, this manuscript with its generated table and figure fragments, and a SHA-256 manifest covering every file — is archived at Zenodo under `[REPLICATION DOI]`. Online Resource 1 supplies a compact reproduction guide containing that manifest, the configuration and provenance summary, a map from each table and figure to the file that produces it, and the archive's identifier. Reproducing every number reported here requires only the analysis scripts and the frozen artifacts, not a re-run of the simulation.

The estimators and the spacing scan are those of the `heavytails` package version 0.5.0, archived at `https://doi.org/10.5281/zenodo.22045594` (Ribeiro, 2026). That record is the software citation; it is not the replication archive for this paper.

A Pre-Asymptotic Stress Test at $n = 10^3$

The frozen design includes $n = 10^3$ explicitly as a pre-asymptotic stress test, because at that sample size $D_n > R_n$ and the intended ordering of the three scales fails. Section 12 therefore restricts its detection and estimator comparisons to the two larger sample sizes, where the ordering holds. The clean control of Table 1 and the intervention check of Table 3 do include $n = 10^3$, because neither depends on how D_n and R_n are ordered. The rest of the pre-specified row is reported here rather than dropped.

Table 9 shows what breaks down, and it is not the theory. With $k_n = \lfloor n^{1/2} \rfloor = 31$, the intervention scale $I^T_{r,k} = \gamma\{rk_n/(k_n - r)\}^{1/2}$ grows quickly in r, and at $r = 10$ it exceeds the detection and risk grid points themselves: the $0.5D$ cell has $S/I^T_{r,k} = 0.803$ and the $0.5R$ cell has $S/I^T_{r,k} = 0.725$. Those cells carry labels defined as fractions of D_n and R_n but in fact lie below $I^T_{r,k}$, so at $n = 10^3$ the frozen signal grid no longer brackets the three scales in the intended order. The clearly negative oracle benefits there are exactly what Proposition 3 predicts below the intervention scale. Across the whole frozen run, at all three sample sizes, every clearly negative oracle benefit occurs in a cell with $S < I^T_{r,k}$.

The detection behavior degrades in the same way and for the same reason. At $1.5D$ detection is 0.951–0.988 rather than 1.0000, and the clean-control no-trim rate falls to 0.9768 $[0.9746, 0.9787]$.

S	r	$S/I^T_{r,k}$	S/D_n	S/R_n	detection [95%]	$\widehat{B}_n^{(A)}$	$\widehat{B}_n^{(T)}$
$0.5I$	1	0.500	0.082	0.091	0.0238 $[0.0199, 0.0284]$	$(-2.51 \pm 0.44)10^{-4}$	$(-2.43 \pm 0.47)10^{-4}$
$0.5I$	3	0.500	0.148	0.164	0.0196 $[0.0161, 0.0238]$	$(-2.45 \pm 0.48)10^{-4}$	$(-6.90 \pm 0.86)10^{-4}$
$0.5I$	5	0.500	0.198	0.219	0.0226 $[0.0188, 0.0271]$	$(-1.41 \pm 0.41)10^{-4}$	$(-1.11 \pm 0.11)10^{-3}$
$0.5I$	10	0.500	0.311	0.345	0.0246 $[0.0207, 0.0293]$	$(-1.91 \pm 0.55)10^{-4}$	$(-2.88 \pm 0.19)10^{-3}$
$1.5I$	1	1.500	0.247	0.274	0.0298 $[0.0254, 0.0349]$	$(-2.41 \pm 0.58)10^{-4}$	$(2.93 \pm 0.73)10^{-4}$
$1.5I$	3	1.500	0.443	0.491	0.0476 $[0.0420, 0.0539]$	$(-3.46 \pm 0.68)10^{-4}$	$(1.06 \pm 0.13)10^{-3}$
$1.5I$	5	1.500	0.594	0.658	0.0896 $[0.0820, 0.0978]$	$(-6.51 \pm 0.86)10^{-4}$	$(2.11 \pm 0.18)10^{-3}$
$1.5I$	10	1.500	0.934	1.035	0.4134 $[0.3998, 0.4271]$	$(-1.82 \pm 0.19)10^{-3}$	$(4.71 \pm 0.28)10^{-3}$
$0.5D$	1	3.034	0.500	0.554	0.0658 $[0.0593, 0.0730]$	$(-5.63 \pm 0.75)10^{-4}$	$(2.27 \pm 0.14)10^{-3}$
$0.5D$	3	1.692	0.500	0.554	0.0670 $[0.0604, 0.0743]$	$(-5.41 \pm 0.85)10^{-4}$	$(1.69 \pm 0.14)10^{-3}$
$0.5D$	5	1.263	0.500	0.554	0.0568 $[0.0507, 0.0636]$	$(-4.24 \pm 0.75)10^{-4}$	$(1.45 \pm 0.17)10^{-3}$
$0.5D$	10	0.803	0.500	0.554	0.0558 $[0.0498, 0.0625]$	$(-5.33 \pm 0.81)10^{-4}$	$(-1.51 \pm 0.22)10^{-3}$
$1.5D$	1	9.101	1.500	1.662	0.9878 $[0.9844, 0.9905]$	$(2.09 \pm 0.04)10^{-2}$	$(2.20 \pm 0.04)10^{-2}$
$1.5D$	3	5.076	1.500	1.662	0.9812 $[0.9770, 0.9846]$	$(1.96 \pm 0.04)10^{-2}$	$(2.10 \pm 0.04)10^{-2}$
$1.5D$	5	3.789	1.500	1.662	0.9758 $[0.9712, 0.9797]$	$(1.94 \pm 0.04)10^{-2}$	$(2.09 \pm 0.04)10^{-2}$
$1.5D$	10	2.408	1.500	1.662	0.9506 $[0.9442, 0.9563]$	$(1.66 \pm 0.04)10^{-2}$	$(1.85 \pm 0.04)10^{-2}$
$0.5R$	1	2.739	0.451	0.500	0.0504 $[0.0447, 0.0568]$	$(-3.73 \pm 0.66)10^{-4}$	$(1.58 \pm 0.12)10^{-3}$
$0.5R$	3	1.528	0.451	0.500	0.0468 $[0.0413, 0.0530]$	$(-3.41 \pm 0.64)10^{-4}$	$(1.29 \pm 0.13)10^{-3}$
$0.5R$	5	1.140	0.451	0.500	0.0506 $[0.0449, 0.0570]$	$(-3.55 \pm 0.78)10^{-4}$	$(6.42 \pm 1.57)10^{-4}$
$0.5R$	10	0.725	0.451	0.500	0.0442 $[0.0388, 0.0503]$	$(-4.65 \pm 0.75)10^{-4}$	$(-1.93 \pm 0.20)10^{-3}$
$1.5R$	1	8.216	1.354	1.500	0.9468 $[0.9402, 0.9527]$	$(1.48 \pm 0.03)10^{-2}$	$(1.81 \pm 0.03)10^{-2}$
$1.5R$	3	4.583	1.354	1.500	0.9422 $[0.9354, 0.9483]$	$(1.41 \pm 0.03)10^{-2}$	$(1.74 \pm 0.04)10^{-2}$
$1.5R$	5	3.421	1.354	1.500	0.9302 $[0.9228, 0.9369]$	$(1.29 \pm 0.03)10^{-2}$	$(1.61 \pm 0.03)10^{-2}$
$1.5R$	10	2.174	1.354	1.500	0.8890 $[0.8800, 0.8974]$	$(1.09 \pm 0.03)10^{-2}$	$(1.43 \pm 0.04)10^{-2}$

Table 9: Pre-asymptotic stress test at $n = 10^3$, the pre-specified row where $D_n > R_n$ and the intended ordering of the scales does not hold. Detection is $\mathbb{P}(\widehat{r} > 0)$ with a Wilson 95% interval; $\widehat{B}_n^{(A)}$ and $\widehat{B}_n^{(T)}$ are the paired adaptive and oracle benefits against contaminated Hill. The $S/I^T_{r,k}$ column shows why this row is a stress test: with $k_n = 31$ and $r = 10$ the intervention scale exceeds the detection and risk grid points ($S/I^T_{r,k} = 0.80$ at $0.5D$ and 0.72 at $0.5R$), so the frozen signal grid no longer brackets the three scales in the intended order. The clearly negative oracle benefits in those cells are what Proposition 3 predicts below $I^T_{r,k}$, not a failure of the oracle.

Both are consequences of $k_n = 31$: the scan has few spacings to work with and the level $\alpha_n = \min\{0.05, 1/\log^2 n\}$ is at its least conservative here. Nothing in this row contradicts the asymptotic statements; it shows the sample size at which the design's own scale separation has not yet appeared.

n	S	r cells	β=0.5	β=1	β=2
10^3	$0.5I$	4	0	0	0
10^3	$1.5I$	4	4	3	2
10^3	$0.5D$	4	3	3	0
10^3	$1.5D$	4	0	0	0
10^3	$0.5R$	4	3	3	0
10^3	$1.5R$	4	1	2	2
10^4	$0.5I$	4	0	0	0
10^4	$1.5I$	3	2	0	0
10^4	$1.5I$=$0.5R$	1	1	1	0
10^4	$0.5D$	4	4	0	0
10^4	$1.5D$	4	0	0	0
10^4	$0.5R$	3	3	3	0
10^4	$1.5R$	4	0	0	0
5×10^4	$0.5I$	4	0	0	0
5×10^4	$1.5I$	4	2	1	0
5×10^4	$0.5D$	4	4	0	0
5×10^4	$1.5D$	4	0	0	0
5×10^4	$0.5R$	4	4	0	0
5×10^4	$1.5R$	4	0	0	0

Table 10: Bounded influence against adaptive trimming across the whole frozen grid. For each design point, the last three columns count the contamination counts $r \in \{1, 3, 5, 10\}$ at which that harmonic-moment member has the larger paired benefit against contaminated Hill. This is the evidence behind the family-level statements of Section 12, which Tables 4 and 5 report at $r = 5$ only. At $n \geq 10^4$ the most aggressively bounded member, $\beta = 2$, never beats adaptive trimming; the four cells where it does are all in the pre-asymptotic $n = 10^3$ row.

n	S	r	S/D_n	β=0.5	β=1	β=2
10^4	$0.5I$	1	0.075	−5.9	−11.0	−16.9
10^4	$0.5I$	3	0.130	−5.5	−11.3	−18.7
10^4	$0.5I$	5	0.170	−6.2	−12.1	−18.9
10^4	$0.5I$	10	0.247	−3.4	−9.0	−16.5
10^4	$1.5I$	1	0.224	−4.0	−10.8	−18.3
10^4	$1.5I$	3	0.391	1.0	−6.4	−15.4
10^4	$1.5I$	5	0.510	6.1	−2.0	−12.7
10^4	$0.5D$	1	0.500	9.2	−0.3	−11.9
10^4	$0.5D$	3	0.500	8.2	−0.1	−10.9
10^4	$0.5D$	5	0.500	6.7	−1.2	−11.9
10^4	$0.5D$	10	0.500	4.3	−1.7	−11.2
10^4	$1.5D$	1	1.500	−5.2	−10.5	−17.4
10^4	$1.5D$	3	1.500	−10.7	−10.1	−17.9
10^4	$1.5D$	5	1.500	−13.4	−9.3	−16.2
10^4	$1.5D$	10	1.500	−16.9	−10.4	−13.6
10^4	$0.5R$	1	0.741	22.0	10.0	−3.9
10^4	$0.5R$	3	0.741	21.3	10.8	−3.4
10^4	$0.5R$	5	0.741	19.4	10.2	−3.6
10^4	$1.5R$	1	2.224	−6.3	−11.3	−17.9
10^4	$1.5R$	3	2.224	−14.5	−9.9	−17.3
10^4	$1.5R$	5	2.224	−20.3	−11.5	−16.6
10^4	$1.5R$	10	2.224	−27.1	−15.1	−14.8
$5{\times}10^4$	$0.5I$	1	0.071	−7.6	−12.5	−18.7
$5{\times}10^4$	$0.5I$	3	0.123	−5.9	−11.5	−18.8
$5{\times}10^4$	$0.5I$	5	0.160	−6.8	−12.4	−19.4
$5{\times}10^4$	$0.5I$	10	0.229	−5.9	−11.3	−18.1
$5{\times}10^4$	$1.5I$	1	0.213	−4.0	−10.3	−18.1
$5{\times}10^4$	$1.5I$	3	0.370	−1.8	−8.4	−16.5
$5{\times}10^4$	$1.5I$	5	0.480	0.3	−6.4	−14.9
$5{\times}10^4$	$1.5I$	10	0.687	8.8	1.4	−8.8
$5{\times}10^4$	$0.5D$	1	0.500	2.1	−5.9	−15.4
$5{\times}10^4$	$0.5D$	3	0.500	1.4	−6.3	−15.6
$5{\times}10^4$	$0.5D$	5	0.500	1.6	−5.7	−14.8
$5{\times}10^4$	$0.5D$	10	0.500	0.2	−6.5	−15.4
$5{\times}10^4$	$1.5D$	1	1.500	−7.8	−13.2	−19.6
$5{\times}10^4$	$1.5D$	3	1.500	−7.7	−10.9	−17.7
$5{\times}10^4$	$1.5D$	5	1.500	−8.9	−11.2	−19.0
$5{\times}10^4$	$1.5D$	10	1.500	−9.4	−9.3	−16.9
$5{\times}10^4$	$0.5R$	1	1.057	5.0	−4.4	−15.0
$5{\times}10^4$	$0.5R$	3	1.057	6.3	−1.2	−12.0
$5{\times}10^4$	$0.5R$	5	1.057	4.8	−1.6	−12.8
$5{\times}10^4$	$0.5R$	10	1.057	3.4	−0.1	−10.9
$5{\times}10^4$	$1.5R$	1	3.170	−7.4	−12.6	−19.0
$5{\times}10^4$	$1.5R$	3	3.170	−8.3	−10.3	−17.9
$5{\times}10^4$	$1.5R$	5	3.170	−14.5	−10.6	−17.7
$5{\times}10^4$	$1.5R$	10	3.170	−22.2	−12.7	−17.5

Table 11: Paired contrasts of every frozen bounded-influence member against adaptive trimming, at every contamination count of the grid. Entries are paired t statistics for the replicate loss differences $d_i = (\widehat{\gamma}^{\star}_{A,i} - \gamma)^2 - (\widehat{\gamma}^{\star}_{\beta,i} - \gamma)^2$, so $t > 0$ favours bounded influence. Table 5 is the $r = 5$ block of this table and Table 10 its sign pattern; the magnitudes are given here so that the family-level statements of Section 12 can be checked cell by cell.

References

J. Beirlant, G. Dierckx, A. Guillou, and C. Stărică. On exponential representations of log-spacings of extreme order statistics. *Extremes*, 5(2):157–180, 2002. https://doi.org/10.1023/A:1022171205129

J. Beran and D. Schell. On robust tail index estimation. *Computational Statistics & Data Analysis*, 56(11):3430–3443, 2012. https://doi.org/10.1016/j.csda.2010.05.028

J. Beran, D. Schell, and M. Stehlik. The harmonic moment tail index estimator: asymptotic distribution and robustness. *Annals of the Institute of Statistical Mathematics*, 66(1):193–220, 2014. https://doi.org/10.1007/s10463-013-0412-2

S. Bhattacharya, M. Kallitsis, and S. Stoev. Data-adaptive trimming of the Hill estimator and detection of outliers in the extremes of heavy-tailed data. *Electronic Journal of Statistics*, 13(1):1872–1925, 2019. https://doi.org/10.1214/19-EJS1561

V. Brazauskas and R. Serfling. Robust and efficient estimation of the tail index of a single-parameter Pareto distribution. *North American Actuarial Journal*, 4(4):12–27, 2000. https://doi.org/10.1080/10920277.2000.10595935

V. Brazauskas and R. Serfling. Small sample performance of robust estimators of tail parameters for Pareto and exponential models. *Journal of Statistical Computation and Simulation*, 70(1):1–19, 2001. https://doi.org/10.1080/00949650108812103

M. Brzezinski. Robust estimation of the Pareto tail index: a Monte Carlo analysis. *Empirical Economics*, 51:1–30, 2016. https://doi.org/10.1007/s00181-015-0989-9

L. de Haan and A. Ferreira. *Extreme Value Theory: An Introduction*. Springer, New York, 2006. https://doi.org/10.1007/0-387-34471-3

L. de Haan and U. Stadtmuller. Generalized regular variation of second order. *Journal of the Australian Mathematical Society*, 61:381–395, 1996. https://doi.org/10.1017/S144678870000046X

D. J. Dupuis and M.-P. Victoria-Feser. A robust prediction error criterion for Pareto modelling of upper tails. *The Canadian Journal of Statistics*, 34(4):639–658, 2006. https://doi.org/10.1002/cjs.5550340406

M. Finkelstein, H. G. Tucker, and J. A. Veeh. Pareto tail index estimation revisited. *North American Actuarial Journal*, 10(1):1–10, 2006. https://doi.org/10.1080/10920277.2006.10596236

B. M. Hill. A simple general approach to inference about the tail of a distribution. *The Annals of Statistics*, 3(5):1163–1174, 1975. https://doi.org/10.1214/aos/1176343247

P.-H. Hsieh. Robustness of tail index estimation. *Journal of Computational and Graphical Statistics*, 8(2):318–332, 1999. https://doi.org/10.1080/10618600.1999.10474816

P. Jordanova, Z. Fabian, P. Hermann, et al. Weak properties and robustness of t-Hill estimators. *Extremes*, 19:591–626, 2016. https://doi.org/10.1007/s10687-016-0256-2

R. Minkah, T. de Wet, and A. Ghosh. Robust estimation of Pareto-type tail index through an exponential regression model. *Communications in Statistics — Theory and Methods*, 52(2):479–498, 2023. https://doi.org/10.1080/03610926.2021.1916530

A. Renyi. On the theory of order statistics. *Acta Mathematica Academiae Scientiarum Hungaricae*, 4:191–231, 1953. `https://doi.org/10.1007/BF02127580`

D. Ribeiro. heavytails: a Python library for heavy-tailed probability distributions, version 0.5.0. Zenodo, 2026. `https://doi.org/10.5281/zenodo.22045594`

P. Ruckdeschel and N. Horbenko. Robust estimators in generalized Pareto models. *Statistics*, 47(4):762–791, 2013. `https://doi.org/10.1080/02331888.2011.628022`

B. Vandewalle, J. Beirlant, A. Christmann, and M. Hubert. A robust estimator for the tail index of Pareto-type distributions. *Computational Statistics & Data Analysis*, 51:6252–6268, 2007. `https://doi.org/10.1016/j.csda.2007.01.003`